TERNARY POLYNOMIAL RELATIONS AND PERIODS OF CUSP FORMS

VASILY ILIN AND KEN ONO

Abstract. Bachmann and van Ittersum conjectured that every polynomial $f \in \mathbb{Q}[x, y, z]$ satisfying

$$f(x, y, z) = f(y, z, x), \qquad f(x, y, z) + f(-y, x - y, z) + f(y - x, -x, z) = 0$$

is a sum of polynomials in two variables. We prove this conjecture over every field of characteristic zero and determine all homogeneous solutions. Moreover, restriction to $z = 0$ identifies the degree n solution space with the space of binary period polynomials of degree n normalized to vanish on both coordinate axes. The solution corresponding to such a polynomial q is

$$f(x, y, z) = q(x, y) + q(y, z) + q(z, x).$$

There are no nonzero homogeneous solutions of odd degree. For even degrees n, the symmetric and alternating solution spaces are each isomorphic over $\mathbb{C}$ to the space of weight $n + 2$ cusp forms on $\mathrm{SL}_2(\mathbb{Z})$. The proof connects the polynomial relations with the cohomology of arithmetic groups through the rational Tits building for GL_3. Cusp forms enter both the description of the solutions and the proof that there are no others. The main theorems and the essential argument are formalized unconditionally in Lean 4.

1. Introduction and statement of results

Classical period polynomials associated with modular forms satisfy linear relations in two variables. Bachmann and van Ittersum considered a three-variable system built from one of these relations [2, p. 481]. Let F be a field of characteristic zero, and suppose that $f \in F[x, y, z]$ satisfies

$$f(x, y, z) = f(y, z, x) \tag{1.1}$$

and

$$f(x, y, z) + f(-y, x - y, z) + f(y - x, -x, z) = 0. \tag{1.2}$$

The first identity imposes cyclic symmetry. The second is the binary period relation in x, y, with z fixed. Its substitution $(x, y) \mapsto (-y, x - y)$ has order three. We denote the sum over its three iterates by Nf, the *three-term norm*. Cyclic symmetry gives the same relation in the other two pairs of variables.

Bachmann and van Ittersum conjectured that every rational solution is a sum of polynomials in two variables. Equivalently, no monomial of a solution involves all three variables. We prove this conjecture over every field of characteristic zero, and we show that every homogeneous solution of odd degree vanishes.

Theorem 1.1. *Let F be a field of characteristic zero. Every polynomial $f \in F[x, y, z]$ satisfying* (1.1) *and* (1.2) *can be written as*

$$f(x, y, z) = P(x, y) + Q(y, z) + R(z, x), \qquad P, Q, R \in F[X, Y]. \tag{1.3}$$

Equivalently, f satisfies $\partial_x \partial_y \partial_z f = 0$. Every homogeneous solution of odd degree is zero.

In fact, restriction to $z = 0$ determines every solution. Write $F[x, y, z]_n$ and $F[X, Y]_n$ for the homogeneous degree n spaces, and $W_n(F)$ for the degree n solutions. Interchanging x, y preserves the relations, so cyclic symmetry gives the decomposition into symmetric and alternating subspaces

$$W_n(F) = W_n^{\mathrm{sym}}(F) \oplus W_n^{\mathrm{alt}}(F).$$

The reconstruction uses the cyclic sum of a binary polynomial q:

$$\Phi(q)(x, y, z) = q(x, y) + q(y, z) + q(z, x).$$

2020 *Mathematics Subject Classification.* Primary 11F67; Secondary 11F11, 11F75, 13A50, 20E42.
Key words and phrases. period polynomials, cusp forms, arithmetic groups, modular symbols.

Theorem 1.2. *Let $n \geq 0$ be even. A symmetric polynomial $f \in F[x,y,z]_n$ satisfies* (1.2) *if and only if*

$$f = \Phi(q), \tag{1.4}$$

where $q \in F[X,Y]_n$ is symmetric, odd in each variable, and satisfies

$$q(X,Y) + q(-Y, X-Y) + q(Y-X, -X) = 0. \tag{1.5}$$

The binary polynomial is uniquely $q(X,Y) = f(X,Y,0)$.

To describe both symmetry types, we let $\mathscr{P}_n^0(F)$ be the space of degree n homogeneous binary polynomials satisfying

$$\begin{gathered} q(X,0) = q(0,Y) = 0, \qquad q(X,Y) + q(-Y,X) = 0, \\ q(X,Y) + q(-Y,X-Y) + q(Y-X,-X) = 0. \end{gathered} \tag{1.6}$$

These are the *normalized binary period polynomials* [17, Section 2]. For positive even n, subtracting $q(1,0)(X^n - Y^n)$ from a polynomial satisfying the two period relations makes it vanish on both axes without changing those relations. Write $\mathscr{P}_n^{0,\mathrm{ev}}(F)$ and $\mathscr{P}_n^{0,\mathrm{odd}}(F)$ for the subspaces even and odd in X, respectively.

Corollary 1.3. *For every $n \geq 0$, restriction is an F-linear isomorphism*

$$W_n(F) \longrightarrow \mathscr{P}_n^0(F), \qquad f \longmapsto f(X,Y,0). \tag{1.7}$$

Its inverse is $q \mapsto \Phi(q)$. It identifies $W_n^{\mathrm{sym}}(F)$ with $\mathscr{P}_n^{0,\mathrm{odd}}(F)$ and $W_n^{\mathrm{alt}}(F)$ with $\mathscr{P}_n^{0,\mathrm{ev}}(F)$.

The two normalized binary period spaces have a classical interpretation in terms of cusp forms. Let $\mathfrak{H} = \{\tau \in \mathbb{C} : \Im\tau > 0\}$ be the upper half-plane, and let $S_k = S_k(\mathrm{SL}_2(\mathbb{Z}))$ denote the complex vector space of holomorphic cusp forms of weight k for the full modular group. For even $n > 0$ and $h \in S_{n+2}$, define its period polynomial by

$$r_h(X,Y) = \int_0^{i\infty} h(\tau)(\tau Y - X)^n \, d\tau. \tag{1.8}$$

The integral converges by cuspidality. Taking the odd part in X, or normalizing the even part as above, gives maps from S_{n+2} to $\mathscr{P}_n^{0,\mathrm{odd}}(\mathbb{C})$ and $\mathscr{P}_n^{0,\mathrm{ev}}(\mathbb{C})$, respectively. Eichler–Shimura theory states that both maps are complex linear isomorphisms [16, 17, 21].

Corollary 1.4. *For every even $n > 0$, the period maps induce complex linear isomorphisms*

$$W_n^{\mathrm{sym}}(\mathbb{C}) \cong S_{n+2}, \qquad W_n^{\mathrm{alt}}(\mathbb{C}) \cong S_{n+2}.$$

For every characteristic zero field F and every $n \geq 0$, we have

$$\dim_F W_n^{\mathrm{sym}}(F) = \dim_F W_n^{\mathrm{alt}}(F) = \dim_{\mathbb{C}} S_{n+2}. \tag{1.9}$$

All these spaces vanish when $n = 0$ or n is odd. Consequently, the generating function of the dimensions is

$$\sum_{n \geq 0} \dim_F W_n(F) t^n = \frac{2t^{10}}{(1-t^4)(1-t^6)}. \tag{1.10}$$

The classification also determines a universal divisor. Set

$$D(x,y,z) = (x+y+z)(x+y)(y+z)(z+x). \tag{1.11}$$

Corollary 1.5. *Over every characteristic zero field, D divides every solution of* (1.1)–(1.2)*. Up to a nonzero scalar, it is the greatest common divisor of all nonzero solutions.*

Example. The first nonzero solutions occur in degree ten. The two binary polynomials in the original problem [2, p. 481] can be written as

$$\begin{aligned} q_{\mathrm{s}}(X,Y) &= XY(X^2-Y^2)^2(X^2-4Y^2)(4X^2-Y^2), \\ q_{\mathrm{a}}(X,Y) &= X^2Y^2(X^2-Y^2)^3. \end{aligned}$$

Both satisfy (1.6). The first is symmetric and odd in each variable, and the second is alternating and even in each variable. Their cyclic sums $f_{\mathrm{s}} = \Phi(q_{\mathrm{s}})$ and $f_{\mathrm{a}} = \Phi(q_{\mathrm{a}})$ form a basis of $W_{10}(F)$. Over $\mathbb{C}$, the binary polynomials q_{s} and q_{a} are nonzero scalar multiples of the odd and normalized even period polynomials, respectively, of Ramanujan's weight twelve discriminant cusp form $\Delta(\tau)$.

Method of proof. The difficulty is not to construct solutions from binary periods, but to show that there are no others. The norm relation mixes two-variable monomials with those involving all three variables. For example,

$$y^2z \longmapsto (x-y)^2z = x^2z - 2xyz + y^2z. \tag{1.12}$$

Thus, we cannot discard the monomials that we wish to exclude.

The geometric part of the proof takes place on the rational Tits building for GL_3. Roughly speaking, this building organizes the rational subspaces of $\mathbb{Q}^3$ and the ways in which they arise from changes of basis. The functional equations will become compatibility relations among certain distinguished cycles in this building, and the original polynomial f arises by evaluating the resulting functional on one particular cycle.

We now describe this picture concretely. The substitutions in the functional equations are invertible linear changes of variables with integer coefficients, so we are naturally led to ordered bases of $\mathbb{Q}^3$, represented by primitive integral vectors, and to the rational subspaces that they span. The rational Tits building for GL_3 is the graph whose vertices are the rational lines and planes in $\mathbb{Q}^3$, with an edge joining a line to a plane precisely when the line is contained in the plane.

In rank three the building is one-dimensional, hence a graph, and its apartments can be described explicitly. An ordered basis (v_1, v_2, v_3) determines the three lines

$$L_i = \mathbb{Q}v_i$$

and the three planes

$$P_{ij} = \mathbb{Q}v_i + \mathbb{Q}v_j.$$

Each line L_i lies in the two planes obtained by adjoining either of the other two basis vectors; among these six subspaces, these are the only incidences. Their incidence graph is therefore the hexagon

$$L_1 \longrightarrow P_{12} \longrightarrow L_2 \longrightarrow P_{23} \longrightarrow L_3 \longrightarrow P_{31} \longrightarrow L_1.$$

Such an oriented hexagon is called an *apartment*. Changing the basis changes the apartment, and the three-term polynomial relation will correspond to a relation among these apartment cycles.

The congruence condition defining the restricted graph is preserved by a natural subgroup of $\mathrm{GL}_3(\mathbb{Z})$. Let

$$K = \{g \in \mathrm{GL}_3(\mathbb{Z}) : (1,1,1)g \equiv (1,1,1) \pmod 3\}.$$

Thus K consists of the integral changes of basis that preserve the coordinate sum functional modulo three, and it acts on the restricted building.

Proposition 3.1 gives the crucial translation from the original polynomial problem to this geometry. A solution determines a K-equivariant polynomial valued functional on the cycles of the restricted graph, and conversely the defining relations of such a functional are exactly the polynomial relations above. Its value on the apartment associated with the standard basis (e_1, e_2, e_3) is the original polynomial f.

The problem is therefore converted into an extension problem. The functional is initially defined only on the restricted graph, whereas the geometric support argument applies to the full Tits building. Proposition 3.4 shows that if the functional extends to the full building, then its value on the standard apartment is a sum of polynomials in two variables. It remains to construct such an extension in each symmetry and parity case, or to prove that the corresponding solution vanishes. Sections 4 and 5 carry this out.

The first task is the common support argument. To carry it out, we pass from the Tits building to an arithmetic quotient of the space of positive definite quadratic forms in three variables, taken modulo positive scaling. Its Borel–Serre compactification has boundary faces indexed by rational lines and planes, with intersections indexed by flags $L \subset P$. Thus the same incidence geometry that appears in the building reappears at the boundary of the arithmetic quotient.

A functional on the full building gives a top-dimensional homology class on this quotient. We prove that such a class comes from the boundary. By Poincaré–Lefschetz duality, this reduces to the vanishing of the corresponding inner cohomology. An explicit positive lower bound for the relevant Laplacian proves this vanishing in the degrees that occur here; the remaining small degrees are handled directly. It therefore remains to understand how a boundary class evaluates on the coordinate apartment. The contributions from the line and plane faces are controlled by elementary unipotent shears, while the contributions at their intersections are controlled by the coboundary relation joining the two adjacent face cocycles. Both types of

contribution have two-variable support. This proves the support statement for any functional that extends to the full building.

We next turn to the extension problem. For symmetric solutions, finite differences produce an explicit polynomial J_f. Its vanishing is exactly the compatibility condition needed to extract the successive binary periods used to construct the extension. We interpret J_f as a boundary coordinate of a degree two cohomology class attached to the restricted-graph functional. The class already vanishes on the boundary faces corresponding to retained vertices, so only the omitted faces remain to be controlled.

For this purpose we construct closed differential forms from classical cusp forms and sum their translates over the arithmetic quotient. The sums converge in the range needed here. We make the same construction with dual coefficients. Applying Stokes's theorem to the original and dual forms gives two boundary identities; comparing them shows that the plane restrictions determine the line restrictions. In even degree a sign symmetry forces the remaining plane restriction to vanish. In odd degree an explicit nonzero integral, together with the nonvanishing of outer critical modular L-values, shows that a nonzero omitted-plane restriction would produce a nonzero retained-line restriction. This contradicts the vanishing already known there. Hence J_f vanishes in both parities.

Once $J_f = 0$, binary period extraction constructs the symmetric extension. In odd degree one candidate term may initially have a simple pole. Its residue is a binary period polynomial, and the space of residues that can arise is stable under the Hecke operators. Liftability forces one extreme period coefficient to vanish on this space. If the residue space were nonzero, Hecke stability would produce an eigenperiod for which the same coefficient is a nonzero modular L-value. Thus the residue space is zero, the pole disappears, and the extension is polynomial.

The alternating cases are shorter. In even degree, signs in the stabilizers of the omitted vertices and edges force the extension obstruction to vanish. In odd degree we use a separate integral argument modulo four: an alternating integer polynomial of odd degree whose norm is divisible by four must have all coefficients even. Scaling a nonzero rational solution to a primitive integer polynomial would contradict this statement. Restriction to $z = 0$ and cyclic summation then complete the classification.

Sections 2–3 establish the polynomial tools, restricted presentation, and support theorem. Sections 4–5 treat the symmetry types, and Sections 6–7 finish the proof and its corollaries. Appendices A–D prove the restricted presentation, the finite-difference and residue–Hecke comparisons, the analytic boundary comparison, and the reduced norm-kernel description.

The external mathematical inputs are the augmented-frame theorems of Church–Putman [10, Theorems B and C], the well-rounded retract [4], and Borel–Serre duality and compactification [8]. The analytic inputs are the spectral-gap estimate and Kuga's formula [7, Lemma 4.1 and (3.6.1)], the nilmanifold comparison with unipotent coefficients [15, Theorem 1.1], and the Eichler–Shimura and Haberland identities [17, Theorems 2.1 and 3.3]. The residue–Hecke comparison uses Hecke-equivariant Shapiro [5, Theorem 3.1]. The applications, coefficient conventions, and additional comparisons needed here are given in the corresponding sections and appendices.

2. Polynomial preliminaries

We separate solutions by degree and symmetry, establish their vanishing on several planes, and extract binary periods when an explicit finite-difference obstruction vanishes. We also fix the cusp form conventions and prove the L-value nonvanishing used later.

Unless stated otherwise, coefficients are rational. Section 6 extends scalars to characteristic zero fields. Put $V_n = \mathbb{Q}[x, y, z]_n$ and use $\mathbb{Q}[X, Y]_n$ for binary forms of degree n. Abbreviate $W_n = W_n(\mathbb{Q})$ and $\mathscr{P}_n^0 = \mathscr{P}_n^0(\mathbb{Q})$.

Write $\mathrm{GL}_d(R)$ for the invertible $d \times d$ matrices over a commutative ring R, and $\mathrm{SL}_d(R)$ for its determinant one subgroup. For a vector space V, write $\mathrm{Sym}^n(V)$ for its nth symmetric power and V^* for its linear dual over the ground field.

The substitutions in the defining equations give linear operators on V_n:

$$\begin{aligned}(Sf)(x,y,z) &= f(y,x,z), && \text{exchange of the first two variables,}\\ (Cf)(x,y,z) &= f(y,z,x), && \text{cyclic permutation,}\\ (Af)(x,y,z) &= f(-y,x-y,z), && \text{the order three substitution.}\end{aligned} \tag{2.1}$$

Products denote composition, with the rightmost operator applied first. Direct substitution gives

$$(A^2f)(x,y,z) = f(y-x,-x,z), \qquad A^3 = I,$$

where I is the identity operator. Thus the three-term norm introduced above is $N = I + A + A^2$, and the defining equations become

$$Cf = f \qquad \text{and} \qquad Nf = 0. \tag{2.2}$$

Let

$$U_n = \mathbb{Q}[x,y]_n + \mathbb{Q}[y,z]_n + \mathbb{Q}[z,x]_n.$$

This is the span of monomials missing a variable. The support assertion is

$$W_n(\mathbb{Q}) = \{f \in V_n : Cf = f,\ Nf = 0\} \subseteq U_n.$$

On cyclic polynomials, the ± 1 eigenspaces of S are the symmetric and alternating subspaces. The next lemma shows that both projections preserve solutions and settles degrees at most three.

Lemma 2.1. *Homogeneous components of a solution are solutions. Its symmetric and alternating parts $(f + Sf)/2$ and $(f - Sf)/2$ are also solutions. There are no nonzero solutions of degree at most three.*

Proof. Each operator in (2.1) preserves degree. Direct substitution gives $SAS = A^2$ and $SCS = C^{-1}$. Hence S preserves both conditions $Nf = 0$ and $Cf = f$. A three-cycle and a transposition generate the full permutation group, which proves the two symmetry assertions. In degrees zero and one, the cyclic spaces are spanned by 1 and $x + y + z$. Their norms are 3 and $3z$, respectively. In degree two, the basis $x^2 + y^2 + z^2, xy + yz + zx$ has norms $4(x^2 - xy + y^2) + 3z^2, \quad x^2 - xy + y^2$. These are linearly independent. The kernels in degrees zero, one, and two are therefore zero.

In degree three, the symmetric basis $p_3 = x^3 + y^3 + z^3$, $m_{21} = x^2y + x^2z + xy^2 + xz^2 + y^2z + yz^2$, and xyz has norms

$$3z^3, \qquad -(x+y)(2x-y)(x-2y) + 4z(x^2 - xy + y^2), \qquad z(x^2 - xy + y^2),$$

which are linearly independent. The alternating basis $\Delta = (x-y)(x-z)(y-z)$ has norm $3xy(x-y) \neq 0$. □

Period extraction requires the following divisibility. Put

$$w = x + y + z, \qquad c_+ = (x+y)(x+z)(y+z).$$

The factors of c_+ define the *opposite pair planes* $x + y = 0$, $x + z = 0$, and $y + z = 0$. The additional factor w in even degree will reduce period extraction to odd degree. Write N_{ij} for the norm in coordinates i, j, fixing the remaining coordinate. Thus $N_{12} = N$, and symmetry gives $N_{13}f = N_{23}f = 0$.

Lemma 2.2. *A symmetric homogeneous solution f is divisible by c_+. In positive even degree it is divisible by wc_+.*

Proof. Write $f \in V_n$, put $\nu = (-1)^n$, and let $g(s,t) = f(s,-s,t)$. Expansion and symmetry give

$$\begin{aligned}&(N_{23}f)(-s,-t,s) - \nu(N_{23}f)(-s-t,-t,s)\\ &\quad - (N_{23}f)(-s-t,-s,-s) + (N_{12}f)(-s-t,0,s) = \nu\{g(s,t) - g(s+t,t)\}.\end{aligned}$$

Therefore, g is translation-invariant in s over $\mathbb{Q}(t)$, hence constant in s. The norm at $(0,0,t)$ makes it zero. Symmetry gives the three factors of c_+. For even degree, put $h(s,t) = f(s,t,-s-t)$. The identity

$$(N_{12}f)(-s-t,t,s) - (N_{23}f)(-s-2t,-t,s) = h(s,t) - h(s+t,t)$$

makes h independent of s. Since $h(s,t) = h(t,s)$, it is constant. Positive homogeneity gives $h = 0$. The distinct linear factors are relatively prime, which proves the product divisibility. □

2.1. **Binary periods and discrete antiderivatives.** We use both binary period spaces in the classification, but only the symmetric space in period extraction.

We write $\epsilon = 1$ for the symmetric case and $\epsilon = -1$ for the alternating case. A ternary polynomial f has *symmetry* ϵ if it is cyclic and $Sf = \epsilon f$. For binary polynomials, define the three-term norm by

$$N_2 q(X, Y) = q(X, Y) + q(-Y, X - Y) + q(Y - X, -X).$$

For $d \geq 0$, let $\mathcal{B}_d^\epsilon$ be the space of polynomials $q \in \mathbb{Q}[X, Y]_d$ satisfying

$$\begin{gathered} q(Y, X) = \epsilon q(X, Y), \qquad q(-X, Y) = q(X, -Y) = -\epsilon q(X, Y), \\ q(X, 0) = q(0, Y) = 0, \qquad N_2 q = 0. \end{gathered} \tag{2.3}$$

We put $\mathcal{B}_d^\epsilon = 0$ for $d < 0$ and abbreviate $\mathcal{B}_d = \mathcal{B}_d^+$ and $\mathcal{B}_d^0 = \mathcal{B}_d^-$. In the notation of the introduction, these are

$$\mathcal{B}_d = \mathscr{P}_d^{0,\mathrm{odd}}(\mathbb{Q}), \qquad \mathcal{B}_d^0 = \mathscr{P}_d^{0,\mathrm{ev}}(\mathbb{Q}).$$

The parity conditions imply $q(-X, -Y) = q(X, Y)$, so both spaces vanish in odd total degree. For $\epsilon = 1$, the axis conditions follow from oddness in each variable. For $\epsilon = -1$, they impose the normalization discussed in the introduction.

Using the symmetry and parity conditions in (2.3), we can rewrite $N_2 q = 0$ as

$$q(X, Y) - q(X - Y, Y) + \epsilon q(X, X - Y) = 0. \tag{2.4}$$

For $q \in \mathcal{B}_d^\epsilon$ with $d > 0$, define its *polynomial lift* from the cyclic sum $\Phi(q)$ by

$$L(q)(x, y, z) = \frac{\Phi(q)(x, y, z)}{x + y + z} = \frac{q(x, y) + q(y, z) + q(z, x)}{x + y + z}. \tag{2.5}$$

To see that this quotient is a polynomial, substitute $(X, Y) = (x + y, y)$ into (2.4) and use $q(y, x) = \epsilon q(x, y)$ to obtain

$$q(x, y) = q(x + y, y) + q(x, x + y).$$

The parity conditions then give $\Phi(q)(x, y, -x - y) = 0$, so $w = x + y + z$ divides $\Phi(q)$. Thus $L(q)$ is homogeneous of degree $d - 1$ and has symmetry ϵ. We set $L(0) = 0$ and write $L(\mathcal{B}_d^\epsilon)$ for the image of this construction. We use these lifts in period extraction, without assuming that they satisfy the ternary norm equation.

We next introduce the operator that relates finite differences to derivatives. Let

$$B(T) = \frac{T}{e^T - 1} = \sum_{r \geq 0} \frac{B_r}{r!} T^r, \qquad B_1 = -\frac{1}{2}, \tag{2.6}$$

be the generating series for the Bernoulli numbers. For a parameter t independent of X, let $(\tau_t g)(X) = g(X - t)$. Taylor's formula gives $\tau_t = e^{-t\partial_X}$, and hence

$$B(-t\partial_X)(\tau_t - I) = -t\partial_X.$$

Thus the Bernoulli operator converts a finite difference into a derivative. The series acts finitely on polynomials and is invertible, since $B(0) = 1$.

We now define the polynomial obstruction used in the symmetric argument. Let X, Y, Z be independent auxiliary variables, distinct from the source variables x, y, z. For a symmetric polynomial f, set

$$\begin{aligned} \psi_1 &= f(Z - 2Y, -X, X - Y), \\ \psi_2 &= \psi_1 - f(Z - X, X - 2Y, -Y) + f(Z - 2Y, Y - X, X - 2Y), \\ J_f &= ZB(-Y\partial_X)\psi_1 - YB(-Z\partial_X)\psi_2. \end{aligned} \tag{2.7}$$

This is the *Bernoulli obstruction*. Its substitutions come from the graph construction in Section 4, where Proposition 4.2 proves $J_f = 0$ for symmetric solutions. Every source triple has coordinate sum $Z - 3Y$, independent of X. Since the Bernoulli operators differentiate only in X, every symmetric polynomial v satisfies

$$J_{w^r v} = (Z - 3Y)^r J_v \qquad (r \geq 0). \tag{2.8}$$

In particular, if $v = w^2 u$, then $J_v = 0$ if and only if $J_u = 0$.

2.2. **Extracting symmetric binary periods.** The next lemma describes the odd degree symmetric kernel of J. It does not assume the norm equation $Nv = 0$.

Lemma 2.3. *Let $m > 0$ be odd and let $v \in V_m$ be symmetric. Then we have*

$$J_v = 0 \quad \Longleftrightarrow \quad v \in \bigoplus_{j\geq 0} w^{2j} L(\mathcal{B}_{m+1-2j}). \tag{2.9}$$

Every lift on the right is polynomial, and the expression is unique.

Proof. We first read the first order data on $w = 0$, then reconstruct a binary period, and finally divide the remainder by w^2.

The first order data. First, direct substitution gives $J_v(X, 0, Z) = Zv(Z, -X, X)$. Thus $J_v = 0$ implies vanishing on each opposite pair plane, and hence $c_+ \mid v$. Write ∂_i for differentiation in the ith argument of v, and put

$$h(X,Y) = v(-Y, X, Y - X), \quad H(X,Y) = (\partial_1 + \partial_2)v(-Y, X, Y - X), \quad p(X,Y) = -\int_0^X H(s,Y)\, ds.$$

Define

$$C_v(X,Y) = p(X - Y, Y) - p(X,Y) - h(X,Y), \qquad \mathcal{D} = -Y\partial_X, \qquad \tau = e^{\mathcal{D}}.$$

The operator τ translates X to $X - Y$. We claim that

$$J_v(X, Y, 3Y) = 0, \qquad \partial_Z J_v|_{Z=3Y} = \tau^2 B(3\mathcal{D})B(\mathcal{D})C_v. \tag{2.10}$$

Here all power series are finite operators on the fixed polynomial degree. To verify the claim, let $v_i = (\partial_i v)(-Y, X, Y - X)$ and set $\psi_i^0 = \psi_i|_{Z=3Y}$, $\psi_i^1 = (\partial_Z \psi_i)|_{Z=3Y}$. Odd homogeneity and symmetry give

$$\psi_1^0 = -h, \quad \psi_2^0 = -(1 + \tau + \tau^2)h, \qquad \psi_1^1 = v_1, \quad \psi_2^1 = (1 + \tau)v_1 - \tau^2 v_3.$$

The chain rule also gives $h_X = v_2 - v_3$, $h_Y = -v_1 + v_3$, and $p_X = -(v_1 + v_2)$. Differentiate the defining expression for J_v, including its Bernoulli operator. The required cancellations are exactly

$$B(3t)(1 + e^t + e^{2t}) = 3B(t),$$
$$-B(t) + tB'(3t)(1 + e^t + e^{2t}) + te^{2t}B(3t) = -e^{2t}B(3t)B(t),$$
$$B(\mathcal{D})(\tau - 1) = \mathcal{D}.$$

The first two identities follow by substituting $B(t) = t/(e^t - 1)$ and clearing denominators. The third is its defining finite-difference identity. This proves (2.10). The operator on its right is invertible, since each factor has constant term one. Consequently $J_v = 0$ gives $C_v = 0$.

Reconstruction of the binary period. Symmetry and homogeneity give

$$h_X = H(X,Y) - H(Y - X, Y), \qquad H(-Y, -X) = H(X,Y). \tag{2.11}$$

Differentiate $C_v = 0$ in X and use the first identity to obtain $H(-X,Y) = H(X,Y)$. Since H has even total degree, it is separately even and symmetric. Thus p is odd in X and even in Y, and $p_X(X,Y) = p_X(Y,-X)$. Define

$$q(X,Y) = -\frac{Xp(Y,-X) + Yp(X,Y)}{m+1}. \tag{2.12}$$

Euler's identity for the degree m polynomial p gives $q_Y = -p$. The displayed formula and the parities of p show that q is symmetric, odd in each variable, and zero on both axes. Since $c_+ \mid v$, we have $h(0,Y) = 0$. Evaluating $C_v = 0$ at $X = 0$ therefore gives $p(Y,Y) = 0$, and hence $q(Y,Y) = 0$.

Set

$$D_q = q(X,Y) - q(X - Y, Y) + q(X, X - Y).$$

Using $q_Y = -p$, symmetry of q, and oddness of p in its first variable, we obtain

$$(\partial_X + \partial_Y)D_q = -p(Y,X) - p(X,Y) + p(X - Y, Y) - p(X - Y, X) = h(X,Y) + h(Y,X) = 0.$$

Here $C_v = 0$ gives the second equality. A homogeneous polynomial of degree $m+1$ annihilated by this derivative is a multiple of $(X-Y)^{m+1}$. At $Y=0$ it equals $q(X,X)=0$. Thus D_q vanishes, which proves $q \in \mathcal{B}_{m+1}$.

The lift and descent in degree. We check directly that every lift has zero obstruction. For $q \in \mathcal{B}_d$, put $t = Z - 2Y$ and $r(X,Y,Z) = q(X,Y) - q(X,t)$. If τ_u translates X to $X-u$, the two frame polynomials of $\Phi(q)$ are

$$((1-\tau_Y)r,\ (1-\tau_Z)r).$$

This follows by expanding the three cyclic sums and using (2.4). Since $B(-u\partial_X)(1-\tau_u) = u\partial_X$, the two terms of $J_{\Phi(q)}$ cancel. The identity $\Phi(q) = wL(q)$ and (2.8) then give $J_{L(q)} = 0$. Cancellation on each opposite pair plane also gives $c_+ \mid L(q)$.

The polynomial extracted from $L(q)$ is the original q. Indeed, differentiate $\Phi(q) = wL(q)$ once in each of the first two variables and restrict to $w=0$. The resulting first-plane data give $p_{L(q)} = -q_Y$, which recovers (2.12). For a symmetric positive degree homogeneous polynomial, $p_v = 0$ is equivalent to $w^2 \mid v$: its three pair sums of first derivatives then vanish on $w=0$, so twice each individual derivative vanishes there. Euler's identity gives $w \mid v$, and differentiation gives a second factor of w. The converse is immediate. It follows that $v - L(q) = w^2 u$. The quotient u is symmetric and satisfies $J_u = 0$ by (2.8). Repeat in degree $m-2$. The final remainder has degree below three and is divisible by c_+, so it is zero. The extracted first-plane data at each step also prove uniqueness. □

The next form of the extraction supplies the two seeds for the symmetric extension. Its only negative exponent occurs in odd degree and accounts for the possible simple pole.

Corollary 2.4. *If a symmetric solution $f \in V_n$ satisfies $J_f = 0$, then uniquely*

$$f = \sum_{j\geq 0} w^{e_j}\Phi(q_j), \qquad \begin{cases} e_j = 2j,\ q_j \in \mathcal{B}_{n-2j}, & n \text{ even}, \\ e_j = 2j-1,\ q_j \in \mathcal{B}_{n+1-2j}, & n \text{ odd}. \end{cases} \tag{2.13}$$

The sum is finite. In odd degree, its first term is interpreted as the polynomial lift $L(q_0)$.

Proof. For $n=0$, the assertion follows from Lemma 2.1. Assume $n>0$. In even degree, Lemma 2.2 gives $f = wv$ with v symmetric of odd degree. Equation (2.8) gives $J_v = 0$. Apply Lemma 2.3 and multiply its expansion by w. In odd degree apply that lemma directly to f. □

2.3. **Classical cusp forms and outer critical values.** We fix the conventions and L-value nonvanishing needed for the boundary comparison and pole removal.

Let $\mathrm{GL}_2^+(\mathbb{R})$ denote the group of real 2×2 matrices with positive determinant. For a holomorphic function h on $\mathfrak{H}$, an integer k, and $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$, define

$$(h|_k A)(\tau) = (\det A)^{k/2}(c\tau + d)^{-k} h\left(\frac{a\tau+b}{c\tau+d}\right).$$

For a finite index subgroup $\Lambda \subset \mathrm{SL}_2(\mathbb{Z})$, a weight k cusp form is holomorphic on $\mathfrak{H}$, satisfies $h|_k\gamma = h$ for $\gamma \in \Lambda$, and has a Fourier expansion with only positive powers at every cusp. This gives exponential decay in each cusp coordinate and the bound

$$(\Im\tau)^{k/2}|h(\tau)| = O(1) \qquad (\tau \in \mathfrak{H}).$$

We now specialize to the full-level space $S_k = S_k(\mathrm{SL}_2(\mathbb{Z}))$. For a positive integer m, define

$$\mathcal{H}_m = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a,b,d \in \mathbb{Z},\ a,d>0,\ ad = m,\ 0 \leq b < d \right\},$$

$$T_m h = m^{k/2-1} \sum_{A \in \mathcal{H}_m} h|_k A.$$

For a prime p, this becomes

$$(T_p h)(\tau) = p^{k-1} h(p\tau) + \frac{1}{p}\sum_{b=0}^{p-1} h\left(\frac{\tau+b}{p}\right).$$

Suppose that h is a simultaneous eigenform for these operators, normalized by

$$h(\tau) = \sum_{m\geq 1} a_h(m) e^{2\pi i m\tau}, \qquad a_h(1) = 1.$$

Then $T_m h = a_h(m)h$, and the Hecke relations give

$$L(h,s) = \sum_{m\geq 1} \frac{a_h(m)}{m^s} = \prod_p L_p(h,s), \qquad L_p(h,s) = \left(1 - a_h(p)p^{-s} + p^{k-1-2s}\right)^{-1}$$

in the common half-plane of absolute convergence. Write $L^{(3)}(h,s) = \prod_{p\neq 3} L_p(h,s)$, omitting the factor at 3. The completed function $\Lambda(h,s) = (2\pi)^{-s}\Gamma(s)L(h,s)$, where Γ is Euler's gamma function, has entire continuation by its Mellin integral. These conventions agree with (1.8) [12, Sections 5.1–5.3 and 5.9–5.10]. The next lemma concerns the last two critical integers in $1,\dots,k-1$.

Lemma 2.5. *If h is a normalized level one cusp eigenform of weight $k > 6$, then $L(h,k-2)$ and $L(h,k-1)$ are nonzero. The same holds after omitting any finite set of Euler factors.*

Proof. The Fourier coefficient integral at height $y = 1/m$ and the bound $y^{k/2}|h(\tau)| \ll 1$ give $a_h(m) = O(m^{k/2})$. Thus the Dirichlet series converges absolutely for $\Re s > k/2 + 1$. To check each local denominator as well, the Hecke recurrence gives

$$\sum_{r\geq 0} a_h(p^r)t^r = (1 - a_h(p)t + p^{k-1}t^2)^{-1}.$$

The coefficient bound makes this holomorphic for $|t| < p^{-k/2}$. Consequently, if the denominator is $(1 - \alpha_p t)(1 - \beta_p t)$, then $|\alpha_p|, |\beta_p| \leq p^{k/2}$. Each local factor is therefore finite and nonzero for $\Re s > k/2$. In the smaller half-plane $\Re s > k/2 + 1$, the sum over primes of the absolute deviations of the local denominators from one converges. Their product is nonzero there. Since $k - 2 > k/2 + 1$ when $k > 6$, both asserted values lie in that half-plane. Removing finitely many nonzero factors preserves nonvanishing. □

3. Arithmetic classes and polynomial support

Proposition 3.1 identifies solutions with polynomial valued functionals on a restricted incidence graph. Proposition 3.4 proves two-variable support for functionals on the full graph. The next two sections construct the required extensions.

3.1. **Cohomology and the restricted incidence graph.** Unless otherwise specified, coefficient modules are rational vector spaces with linear group actions. For a left H-module M, let $C^r(H,M)$ be the functions $H^r \to M$ with the group coboundary, and let $H^r(H,M)$ be its cohomology. We use right modules for group homology $H_r(H,M)$.

The invariant and coinvariant spaces are

$$M^H = \{m \in M : hm = m \text{ for every } h \in H\}, \qquad M_H = M/\langle hm - m : h \in H,\ m \in M\rangle.$$

In particular, $H^0(H,M)$ is the invariant space M^H. In degree one, a cocycle is a *crossed homomorphism*, a map $b : H \to M$ satisfying

$$b(gh) = b(g) + g\,b(h).$$

Two such maps represent the same cohomology class if their difference has the form $g \mapsto gm - m$ for some $m \in M$. In degree two, a cocycle is a function $z : H \times H \to M$ satisfying

$$g\,z(h,k) - z(gh,k) + z(g,hk) - z(g,h) = 0.$$

A two-coboundary has the form $g\,b(h) - b(gh) + b(g)$ for a function $b : H \to M$. These are the formulas used for the connecting class in Section 4 [9, Chapters III–IV].

Write $\mathrm{Hom}_H(V,M)$ for linear maps satisfying $\varphi(hv) = h\varphi(v)$, and $\mathrm{Ext}^1_H(V,M)$ for equivalence classes of extensions $0 \to M \to E \to V \to 0$. For spaces, H_r and H^r denote singular homology and cohomology with the specified local system. A tilde denotes reduced homology. On a simply connected cover the coefficients are constant, with transport under deck transformations given by the group action.

For $1 \to U \to H \to L \to 1$, we use the Hochschild–Serre spectral sequence

$$E_2^{p,q} = H^p(L, H^q(U, M)) \implies H^{p+q}(H, M). \tag{3.1}$$

It computes U-cohomology first, followed by L-cohomology [9, Chapter VII].

Our maximal parabolic groups stabilize rational lines or planes and are block upper triangular in adapted coordinates. The subgroup with identity diagonal blocks is the *unipotent radical*, and the diagonal blocks give the *Levi quotient*. For an arithmetic subgroup, U is a lattice in the radical and L its Levi quotient. A radical element is a *unipotent shear*

$$u(B) = \begin{pmatrix} I_r & B \\ 0 & I_s \end{pmatrix}, \qquad B \in M_{r\times s}(\mathbb{Z}),$$

where I_r, I_s are identity matrices. It acts on the two coordinate blocks by $(x, y) \mapsto (x + By, y)$.

The arithmetic Levi groups occurring here have free subgroups of finite index, so their rational cohomology vanishes above degree one [20, Chapter I]. Finite quotients are handled by averaging over $\mathbb{Q}$. Consequently, only the columns $p = 0, 1$ of (3.1) can contribute in these applications.

We now specify the arithmetic group and its polynomial coefficients. Let $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$, write e_i for the standard coordinate vectors and E_{ij} for the matrix units, and put $U_{ij} = I + E_{ij}$ for $i \neq j$. Set

$$G = \mathrm{GL}_3(\mathbb{Z}), \qquad \ell = (1, 1, 1) \in (\mathbb{F}_3^3)^*, \qquad K = \{g \in G : \ell g = \ell \pmod 3\}.$$

Thus K fixes the covector ℓ modulo three, not merely its line.

For $t = 0, 1$, our coefficient modules are

$$E_{n,t} = \mathrm{Sym}^n((\mathbb{Q}^3)^*) \otimes \overset{-t}{\det}.$$

Their underlying vector space is $V_n = \mathbb{Q}[x, y, z]_n$. The determinant twist changes the action, not the polynomials:

$$g \cdot f(\mathbf{x}) = \det(g)^{-t} f(g^{-1}\mathbf{x}), \qquad \mathbf{x} = (x, y, z)^{\mathrm{t}}.$$

The choices $t = 0$ and $t = 1$ will correspond to the alternating and symmetric cases, respectively.

We distinguish this left action from direct substitution and from the right action used in homology:

$$(f \mid g)(\mathbf{x}) = f(g\mathbf{x}), \qquad f \mid_t g = g^{-1} \cdot f = \det(g)^t (f \mid g).$$

The unindexed bar $\mid$ always denotes direct substitution, without a determinant twist. We use the left action in cohomology and equivariant Hom, and the right action $\mid_t$ in homology. The left action also induces a right action on the dual module:

$$(\Psi \cdot g)(v) = \Psi(g \cdot v).$$

For a rational line or plane $W \subset \mathbb{Q}^3$, consider its integral lattice $W \cap \mathbb{Z}^3$. We call W *good* if this lattice contains a vector whose coordinate sum is nonzero modulo three, and *bad* otherwise. Equivalently, ℓ is nonzero on the image of $W \cap \mathbb{Z}^3$ in $\mathbb{F}_3^3$. For example, $\mathbb{Q}e_1$ is good, whereas $\mathbb{Q}(e_1 - e_2)$ is bad. The condition is imposed separately on each subspace: the good plane $\mathbb{Q}e_1 + \mathbb{Q}e_2$ contains the bad line $\mathbb{Q}(e_1 - e_2)$.

Let $\mathcal{T}$ be the graph whose vertices are the rational lines and planes in $\mathbb{Q}^3$, with an edge joining a line to a plane precisely when the line lies in the plane. This is the rational Tits building for GL_3, a spherical building of type A_2 [1]. Let $\mathcal{T}^{\uparrow}$ be the subgraph consisting of the good vertices and all edges between them. Define

$$\mathrm{St} = \widetilde{H}_1(\mathcal{T}; \mathbb{Q}), \qquad \mathrm{St}^{\uparrow} = \widetilde{H}_1(\mathcal{T}^{\uparrow}; \mathbb{Q}).$$

The first is the *Steinberg module*. For a graph, first homology consists of finite rational edge chains with zero boundary. Thus $\mathrm{St}^{\uparrow} \subset \mathrm{St}$, and both are K-modules.

An ordered basis (v_1, v_2, v_3) of $\mathbb{Q}^3$ determines three lines $L_i = \mathbb{Q}v_i$ and three planes $P_{ij} = \mathbb{Q}v_i + \mathbb{Q}v_j$. The cycle

$$L_1 \longrightarrow P_{12} \longrightarrow L_2 \longrightarrow P_{23} \longrightarrow L_3 \longrightarrow P_{31} \longrightarrow L_1$$

is the *apartment* determined by this ordered basis. Permuting the basis vectors changes its orientation by the sign of the permutation. The apartment of the standard basis (e_1, e_2, e_3) is called the *identity apartment*. All its vertices are good.

The next proposition identifies solutions by evaluation on the identity apartment. Its presentation uses the matrices

$$A = \begin{pmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix},$$

and let S_0 interchange the first two coordinate vectors. Here A and C are the matrix versions of the substitutions in (2.1), with the coefficient actions fixed above. All three matrices lie in K.

The group algebra $\mathbb{Q}K$ records rational linear combinations of translates of the identity apartment. The relations $1 + A + A^2$, $C - 1$, and $S_0 + 1$ encode the norm, cyclic invariance, and the transposition sign. The determinant twist converts the last sign into the required polynomial symmetry.

The analogous presentation for the full incidence graph is due to Bykovskii. Church–Putman give a topological proof using augmented frames [10, Theorem B]. The congruence-restricted presentation requires the additional argument in Appendix A.

Proposition 3.1. *The graph $\mathcal{T}^{\uparrow}$ is connected, and we have*

$$\mathrm{St}^{\uparrow} \cong \frac{\mathbb{Q}K}{\mathbb{Q}K(1 + A + A^2) + \mathbb{Q}K(C - 1) + \mathbb{Q}K(S_0 + 1)}. \tag{3.2}$$

Evaluation on the identity apartment gives isomorphisms

$$\mathrm{Hom}_K(\mathrm{St}^{\uparrow}, E_{n,0}) \cong W_n^{\mathrm{alt}}(\mathbb{Q}), \qquad \mathrm{Hom}_K(\mathrm{St}^{\uparrow}, E_{n,1}) \cong W_n^{\mathrm{sym}}(\mathbb{Q}).$$

Proof. A partial augmented frame is a subset of an integral basis, possibly together with one additional primitive line represented by a signed sum of two basis vectors. Let Y be the full good subcomplex of the complex of partial augmented frames of Church–Putman [10, Theorem C]. It is connected and has $H_1(Y; \mathbb{Q}) = H_2(Y; \mathbb{Q}) = 0$. Appendix A proves the required restriction argument. If Y^0 consists of simplices spanning proper subspaces, pass to the poset of its nonempty simplices, ordered by inclusion. The span map takes this poset to the poset of good proper rational subspaces. The inverse image of the lower interval at a line is a point. At a plane it is the face poset of its good filled Farey complex. These spaces are contractible by Appendix A. Quillen's Theorem A [18], applied to these posets, gives a homotopy equivalence of their order complexes and hence a K-equivariant homology identification $H_1(Y^0; \mathbb{Q}) = \mathrm{St}^{\uparrow}$. The relative complex of (Y, Y^0) starts in degree two. Its generators are good unimodular frames, and its relations are good augmented frames.

Every good primitive line has a unique primitive integral representative of ℓ-value 1 modulo three. Normalized good bases are therefore a single K-orbit. In a good augmented core, reduction modulo three forces all signs in its three-vector relation to agree. The normalized vectors sum to zero. For $r = -e_1 - e_2$, its boundary, modulo proper-span faces, is $\partial[r, e_1, e_2, e_3] = [I] + [A] + [A^2]$. Column permutations give $C - 1$ and $S_0 + 1$. The relative homology sequence now gives (3.2), and also connectedness of Y^0. The determinant twist reverses the transposition condition on the polynomial. Finally $A^{-1} = A^2$, so the left-action norm is the same polynomial norm. □

3.2. **Ordinary cycles.** We call full-building classes *ordinary cycles*. Their finite coefficient model makes apartment evaluation explicit.

Let $\mathscr{X}$ be the space of positive definite ternary real quadratic forms modulo positive scaling. Determinant one representatives give

$$\mathscr{X} \cong \mathrm{SL}_3(\mathbb{R})/\mathrm{SO}(3), \qquad g\mathrm{SO}(3) \longleftrightarrow Q = (gg^{\mathsf{t}})^{-1},$$

where $\mathrm{SO}(3)$ is the group of real orthogonal matrices of determinant one. The space $\mathscr{X}$ has real dimension five. For $g \in G$, we use the left action

$$g \cdot Q = g^{-\mathsf{t}} Q g^{-1}.$$

The corresponding right action is $Q \mapsto g^{\mathsf{t}} Q g$. The analytic appendix uses the Gram matrix $gg^{\mathsf{t}} = Q^{-1}$ rather than Q, so the distinction between these actions matters.

Since K stabilizes ℓ modulo three, its coset space is finite: reduction gives the bijection

$$K\backslash G \xrightarrow{\sim} \mathcal{S} = \mathbb{F}_3^3 \setminus \{0\}, \qquad Kg \longmapsto \ell g \pmod 3.$$

Thus the induced module consists of functions on the 26 nonzero row covectors:

$$F : \mathcal{S} \longrightarrow V_n.$$

For $g \in G$, the right action is

$$(F \mid_t g)(r) = \det(g)^t (F(rg^{-1}) \mid g).$$

Covectors are multiplied modulo three, and the polynomial bar is untwisted.

To determine which lists represent homology classes, we use Ash's well-rounded retract [4]. This is a contractible, three-dimensional cell complex inside $\mathscr{X}$. Its points are the classes of forms whose shortest nonzero integral vectors span $\mathbb{R}^3$. For this construction we choose minimum-one representatives, rather than determinant one representatives:

$$\min_{0 \neq v \in \mathbb{Z}^3} v^{\mathrm{t}} Q v = 1.$$

Its equivariant cellular complex computes rational group homology [9, Chapter VII, Section 7]. Since there are no cells above degree three, top cycles represent $H_3(K, E_{n,t})$. The retract is three-dimensional, whereas the arithmetic quotient is five-dimensional. Poincaré–Lefschetz duality on its Borel–Serre compactification relates degree three homology to degree two cohomology.

The standard top-dimensional cell is stabilized by the signed permutation matrices. For such a matrix $h = D\pi$, where D is diagonal with entries ± 1 and π is a permutation matrix, define

$$\chi_0(h) = \operatorname{sgn}(\pi), \qquad \chi_1(h) = \prod_i D_{ii}.$$

The full apartment presentation [10, Theorem B] gives the following dual model, agreeing with Allison–Ash–Conrad [3]. Identity apartment evaluation is $F(\ell)$.

Lemma 3.2. *The group $H_3(K, E_{n,t})$, with the right coefficient action defined above, is represented by precisely the functions $F : \mathcal{S} \to V_n$ such that*

$$F(rh) = \chi_t(h) F(r) \mid h, \tag{3.3}$$

$$F(r) + F(rA^2) \mid A + F(rA) \mid A^2 = 0, \tag{3.4}$$

for every $r \in \mathcal{S}$ and every signed permutation matrix h. Every bar in (3.3)–(3.4) *is untwisted. The determinant factor has already been included in χ_t. The coordinate $F(\ell)$ is the value on the identity apartment of the corresponding ordinary Steinberg functional.*

Proof. Let $E = E_{n,t}$. Rational Borel–Serre duality gives

$$H^3(K, E^*) \cong (\mathrm{St} \otimes E^*)_K, \qquad H_3(K, E) \cong \mathrm{Hom}_K(\mathrm{St}, E),$$

where the second isomorphism is obtained by dualizing the first, with the coefficient conventions fixed above. The virtual cohomological dimension is three, and the orientation character is trivial [8]. Induction over the finite coset space $K\backslash G$ identifies the latter space with G-equivariant maps from St to the induced coefficient module [9, Chapter III, Section 5].

The full apartment presentation [10, Theorem B] has signed-permutation relations and the relation $1 + A + A^2$. The first says that $F \mid_t h = \operatorname{sgn}(\pi) F$, which is (3.3). The second says that $F + F \mid_t A + F \mid_t A^2 = 0$, which is (3.4) after evaluating at r. These relations are sufficient by the presentation theorem. Under induction, the identity coset is indexed by ℓ, so evaluation there recovers the original apartment value. Lemma 3.3 fixes the geometric normalization of this evaluation. □

The cellular description also applies to arbitrary rational coefficient modules: averaging over finite cell stabilizers is exact, and the retract has no cells above degree three. In particular, for the full group G with determinant-twisted polynomial or rational function coefficients, a top-cycle coefficient is symmetric, odd in each variable, and annihilated by N. We use this generality in Section 4.4, where the coefficient module is enlarged to allow simple poles.

3.3. **Apartment evaluation and boundary duality.** For an ordinary class z, the polynomial of interest is $F_z(\ell)$. We express this evaluation as a boundary pairing.

The Borel–Serre bordification $\overline{\mathscr{X}}$ adds faces indexed by rational parabolic subgroups [8]. Maximal faces correspond to lines and planes, and their intersections to flags $L \subset P$. Its boundary is homotopy equivalent to the rational Tits building, so apartments have geometric representatives there.

To handle torsion, pass to a neat normal congruence subgroup: for each of its elements, the subgroup of $\mathbb{C}^\times$ generated by the eigenvalues contains no nontrivial root of unity. Such a subgroup exists, and its quotient of $\overline{\mathscr{X}}$ is a compact manifold with corners. After the geometric constructions, take coinvariants in homology and invariants in cohomology under the finite covering group. Rational averaging is exact. We write $\overline{\mathscr{X}}_G$ and $\partial\overline{\mathscr{X}}_G$ for these equivariant quotients, without assuming that the quotient by G itself is a manifold.

Let

$$M = \operatorname{Ind}_K^G E_{n,t}$$

with the preceding coefficient actions, and let $\mathcal{M}$ be its local system. Shapiro's lemma gives

$$H_3(\overline{\mathscr{X}}_G; \mathcal{M}) \cong H_3(K, E_{n,t}).$$

The top-cell coefficient $F_z \in M$ of Lemma 3.2 is determined by z, since the retract has no degree four cells. Its value $F_z(\ell)$ records the identity apartment.

To express evaluation as a scalar pairing, choose $\varphi \in M^*$. The determinant one diagonal forms

$$\operatorname{diag}(a, b, (ab)^{-1}), \qquad a, b > 0,$$

form a two-dimensional flat in $\mathscr{X}$. Its compactified image defines the *coordinate modular symbol* [6]. With constant coefficient φ on the universal flat, it gives a relative homology class

$$a_\varphi \in H_2(\overline{\mathscr{X}}_G, \partial\overline{\mathscr{X}}_G; \mathcal{M}^*).$$

Under the homotopy equivalence between the universal boundary and the rational Tits building, the boundary of the flat represents the coordinate apartment. We orient the flat and the averaged standard top-cell generator so that their local intersection number is $+1$. With this normalization, the next lemma compares evaluation of a boundary class with its boundary intersection pairing. We use Poincaré–Lefschetz duality with the indicated local coefficients [14, Sections 3.3 and 3.H].

Lemma 3.3. *Let j be the inclusion of the boundary, and let $\mathrm{PD}_\partial : H_3(\partial\overline{\mathscr{X}}_G; \mathcal{M}) \to H^1(\partial\overline{\mathscr{X}}_G; \mathcal{M})$ be boundary Poincaré duality. The diagram*

$$\begin{array}{ccc} H_3(\partial\overline{\mathscr{X}}_G; \mathcal{M}) & \xrightarrow{j_*} & H_3(\overline{\mathscr{X}}_G; \mathcal{M}) \\ \Big\downarrow{\scriptstyle \mathrm{PD}_\partial} & & \Big\downarrow{\scriptstyle z \mapsto \varphi(F_z)} \\ H^1(\partial\overline{\mathscr{X}}_G; \mathcal{M}) & \xrightarrow{\omega \mapsto \langle \omega, \partial a_\varphi \rangle} & \mathbb{Q} \end{array} \tag{3.5}$$

commutes. In particular, for $\lambda \in V_n^*$ and $\varphi_\lambda(F) = \lambda(F(\ell))$, we have

$$\lambda\big(F_{j_* c}(\ell)\big) = \big\langle \mathrm{PD}_\partial\, c, \partial a_{\varphi_\lambda} \big\rangle. \tag{3.6}$$

This is compatible with induction of coefficients and with passage to a finite index subgroup.

Proof. Intersection with the flat reads the top-cell coefficient. Moving the intersection into a boundary collar gives the bottom arrow.

For the first, normalize a form in the standard open top cell by requiring that e_1, e_2, e_3 have length one. Near the identity matrix its Gram matrix is

$$\begin{pmatrix} 1 & r_{12} & r_{13} \\ r_{12} & 1 & r_{23} \\ r_{13} & r_{23} & 1 \end{pmatrix}.$$

The diagonal flat meets this cell at the identity, transversely. It meets no other open top cell: a diagonal form is well rounded only when all three diagonal entries agree, and at that form the shortest primitive

vectors are precisely $\pm e_i$. The off-diagonal coordinates orient the cell, and the two diagonal ratios orient the flat. A permutation has its permutation sign on the cell, while a coordinate sign change has sign $+1$, exactly as in the top-cell model. The local intersection therefore evaluates the coefficient by φ.

Translate this calculation by an integral basis matrix. The cellular orientation and the dual coefficient transport are respectively the signed permutation character and $\varphi \cdot g$. Thus the local intersections assemble with the same relations as the modular symbol pairing [3, Section 3, (3–1)]. With induced coefficients, unfolding a chain over $K\backslash G$ labels the translated basis by its covector ℓg. The functional φ_λ selects the identity coset, not the sum of all covectors. Hence the intersection of z with a_φ is $\varphi(F_z)$.

For the second comparison, smooth the neat cover's corners and move c slightly inward along a collar, transverse to the relative flat. The intersections are those of c with the flat's boundary. With the boundary orientation induced by this collar, the cap-product definition gives $\langle \mathrm{PD}_\partial\, c, \partial a_\varphi \rangle$. The identity holds on chains modulo boundaries by naturality of Poincaré–Lefschetz duality [8].

Both constructions commute with coefficient transport and subdivision. Coset unfolding is the cellular Shapiro map, proving compatibility with induction. Taking finite group coinvariants in homology and invariants in cohomology is exact and preserves the pairing, giving descent to the stated level. □

3.4. **The support principle.** The pairing in Lemma 3.3 now gives support for both coefficient systems. The remaining task will be extension.

Proposition 3.4. *If $n > 0$, every K-equivariant map $\mathrm{St} \to E_{n,t}$ takes the identity apartment into U_n, for $t = 0, 1$.*

Proof. Passage to the boundary. Choose a neat normal subgroup $\Gamma' \subset K \cap \mathrm{SL}_3(\mathbb{Z})$ and put $X_{\Gamma'} = \Gamma'\backslash \mathscr{X}$. Proposition C.1, proved in Appendix C, gives

$$H^2_c(X_{\Gamma'}, E_{n,t} \otimes \mathbb{C}) \longrightarrow H^2(X_{\Gamma'}, E_{n,t} \otimes \mathbb{C}) \quad \text{equal to zero.}$$

Extension to $\mathbb{C}$ is faithful, so the rational map is zero as well.

The five-dimensional symmetric space has trivial orientation character: on symmetric matrices $Q \mapsto g^{\mathsf{t}} Q g$ has determinant $(\det g)^4 > 0$, and quotienting by positive scaling removes an oriented radial direction. Poincaré–Lefschetz duality identifies the absolute-to-relative map in degree three homology with the zero map above. Hence boundary homology surjects onto ordinary top homology. Put $\Delta = K/\Gamma'$. Rational Hochschild–Serre and averaging give

$$H_3(K, E_{n,t}) \cong H_3(\Gamma', E_{n,t})_\Delta.$$

The same finite-cover descent applies to boundary homology and commutes with its map to ordinary homology, so exact averaging preserves the surjection at level K. It remains to evaluate an arbitrary boundary class.

Face cocycles and radical invariance. For evaluation, use $M = \mathrm{Ind}_K^G E_{n,t}$ and $\partial = \partial\overline{\mathscr{X}}_G$ as in Lemma 3.3. Shapiro's finite covector factor retains every level three cusp. The two maximal parabolic types for G are represented by the upper integral parabolics P and Q of block types $(2,1)$ and $(1,2)$, with intersection the Borel group B_0.

Take collar neighborhoods of the two unions of closed maximal faces on the neat cover. Their intersections are Borel strata, with no triple intersection because a flag has only a line and a plane. Induction and Shapiro give the Mayer–Vietoris sequence

$$H^0(P,M) \oplus H^0(Q,M) \to H^0(B_0,M) \to H^1(\partial, M) \to H^1(P,M) \oplus H^1(Q,M) \to H^1(B_0,M). \tag{3.7}$$

To keep the corner signs explicit, use the mapping cone

$$\mathcal{C}^r = C^r(P,M) \oplus C^r(Q,M) \oplus C^{r-1}(B_0,M),$$
$$d(a,b,c) = (da, db, \mathrm{Res}\, a - \mathrm{Res}\, b - dc).$$

The dual chain differential is $\partial(x,y,z) = (\partial x + i_P z, \partial y - i_Q z, -\partial z)$, where i_P, i_Q are the two inclusions. The evaluation pairing is the sum of the three component pairings. These differentials are adjoint. In particular

a degree-one cocycle is (A_P, A_Q, u) with $du = \operatorname{Res} A_P - \operatorname{Res} A_Q$. For $R = P$ or Q, let $N_R \cong \mathbb{Z}^2$ be its unipotent radical and L_R its Levi quotient. The five-term sequence shows

$$H^1(R, M) = H^1(L_R, M^{N_R}). \tag{3.8}$$

This means that face cocycles may be chosen to factor through the Levi, with values fixed by the radical. We verify the necessary vanishing. Decompose the covectors into N_R-orbits and apply Shapiro to their stabilizer lattices. Their unipotent coefficient systems satisfy the nilmanifold comparison [15, Theorem 1.1]. Lie and group cohomology therefore agree, naturally for the Levi action, as explained in Appendix C.4. We may compute after complexification, which detects the required rational vanishing.

We now compute the Lie cohomology explicitly. For the root operators $z\partial_x, z\partial_y$, the cocycle equation for a pair (a, b) is $a_y = b_x$. Such a pair is the gradient of a homogeneous potential of degree $n+1$. Exact pairs are the gradients of potentials divisible by z, so the quotient is the binary symmetric power of degree $n+1$. For $x\partial_z, y\partial_z$, subtract an exact pair to remove its common z-dependent part. The remaining binary pairs are taken modulo (xg, yg), and $(a, b) \mapsto xb - ya$ identifies that quotient with binary forms of degree $n+1$. These descriptions cover the standard and dual coefficient conventions.

It remains to take Levi invariants. In either case $H^1(N_R, E_{n,t})$ restricts to $\operatorname{Sym}^{n+1}(\mathbb{Q}^2)$ or its dual on the derived subgroup SL_2 of the Levi. The corresponding level three subgroup fixes every covector. Its two opposite root unipotents have no common invariant in this positive binary degree. Consequently $H^1(N_R, M)^{L_R}$ vanishes, which proves (3.8).

If $v \in M^{N_R}$, the lattice fixing ℓ contains $3N_R$ and is Zariski dense in the algebraic radical. Therefore $v(\ell)$ is invariant under its root derivatives and belongs to U_n. The same conclusion holds for invariants of the common root subgroup of two adjacent maximal radicals.

Evaluation on the coordinate apartment. Let c be a boundary top class and $z_c = F_{j_* c}$ its induced top-cell coefficient. Let $U_n^\perp = \{\lambda \in V_n^* : \lambda|_{U_n} = 0\}$. To prove $z_c(\ell) \in U_n$, take $\lambda \in U_n^\perp$ and put $\Phi_\lambda(v) = \lambda(v(\ell))$. Lemma 3.3 gives

$$\lambda(z_c(\ell)) = \langle \omega, \partial a_{\Phi_\lambda} \rangle, \qquad \omega = \mathrm{PD}_\partial\, c \in H^1(\partial, M).$$

The boundary apartment has successive vertices L_1, P_{12}, L_2, P_{23}, L_3, P_{31}, L_1, where $L_i = \mathbb{Q}e_i$ and $P_{ij} = \mathbb{Q}e_i + \mathbb{Q}e_j$. We evaluate its face and corner terms in the mapping-cone convention above.

By (3.8), a mapping-cone representative of ω is (A_P, A_Q, u), where the face cocycles are inflated and $du = \operatorname{Res} A_P - \operatorname{Res} A_Q$. Here are the coefficient transports on the six edges. For an ordered permutation σ, let $g_\sigma e_i = e_{\sigma(i)}$. Starting from the flag $L_1 \subset P_{12}$, the permutations are

$$(1,2,3), \quad (2,1,3), \quad (2,3,1), \quad (3,2,1), \quad (3,1,2), \quad (1,3,2),$$

with incidence signs $+, -, +, -, +, -$, respectively. The transported dual coefficient is $\Phi_\lambda \cdot g_\sigma$. The left action inside this formula includes the determinant twist. The listed signs are the separate cellular orientation signs. Every g_σ fixes ℓ and preserves U_n. Each transported coefficient therefore annihilates the appropriate radical invariant space, and all face pairings are zero. Further transport within a maximal parabolic preserves that space.

For a corner, the common root group is a translate of $N_0 = \{I + aE_{13} : a \in \mathbb{Z}\}$. Its coefficient functional annihilates M^{N_0}, and finite-dimensional duality gives $(M^{N_0})^\perp = \operatorname{span}\{\Psi \cdot n - \Psi : n \in N_0,\ \Psi \in M^*\}$. Thus the corner zero-chain can be written $r = \sum_i (\Psi_i \cdot n_i - \Psi_i)$, with $n_i \in N_0$. In the right-module bar complex, $\partial(\Psi[n]) = \Psi \cdot n - \Psi$, so $s = \sum_i \Psi_i[n_i]$ has $\partial s = r$. Both face cocycles are inflated from their Levis and vanish on this common radical. Adjointness gives

$$\langle u, r \rangle = \sum_i \Psi_i(n_i \cdot u - u) = \langle du, s \rangle = \langle \operatorname{Res} A_P - \operatorname{Res} A_Q, s \rangle = 0.$$

Hence, the corner pairing vanishes by the coboundary equation for u (no invariance of u is required). We obtain $\lambda(z_c(\ell)) = 0$ for every annihilator λ, so $z_c(\ell) \in U_n$. Lemma 3.2 completes the proof. □

4. Symmetric solutions

Let $f \in V_n$ be a symmetric solution with $n > 0$. Constants are settled by Lemma 2.1. We prove $J_f = 0$, use Corollary 2.4 to construct an extension, and remove its possible odd degree pole by Hecke operators. The support theorem then applies.

4.1. **An obstruction with boundary restrictions.** We realize J_f as a boundary coordinate of a degree two class attached to the restricted-graph functional.

Conjugate the level covector $\ell = (1,1,1)$ to $e_3^* = (0,0,1)$ by

$$R_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}, \qquad B_3 = R_3^{-1}, \qquad K^\dagger = B_3 K B_3^{-1}, \qquad \mathcal{T}^\dagger = B_3 \mathcal{T}^\uparrow.$$

Since $e_3^* B_3 = \ell$, the group $K^\dagger$ fixes e_3^* modulo three. A subspace is now good when its integral lattice contains a vector with last coordinate nonzero modulo three. Write $E_n = E_{n,1}$. Transport of the graph and coefficients gives a $K^\dagger$-equivariant functional

$$F : H_1(\mathcal{T}^\dagger; \mathbb{Q}) \longrightarrow E_n.$$

We fix its polynomial normalization. If V is an ordered integral basis with good column lines, let $\mathcal{A}(V)$ be its oriented apartment. Choose column signs so that their last coordinates are one modulo three, and call the resulting matrix V_c. Then the value of F on this apartment is

$$F(\mathcal{A}(V)) = \det(V_c) f(V_c^{-1}\mathbf{X}), \qquad \mathbf{X} = (X,Y,Z)^{\mathsf{t}}. \tag{4.1}$$

Indeed, $V_c R_3 \in K^\dagger$, and the reference apartment $\mathcal{A}(B_3)$ has value $f(R_3\mathbf{X})$, so equivariance gives this formula.

Choose a good vertex v and oriented paths $\tau(g)$ from v to gv for $g \in K^\dagger$. Their endpoints make

$$z_F(g,h) = F\big(\tau(g) + g\tau(h) - \tau(gh)\big) \tag{4.2}$$

well defined. The path terms cancel in the group coboundary, and changing paths changes z_F by a coboundary. Thus $a_F = [z_F] \in H^2(K^\dagger, E_n)$ is the connecting class [9, Chapter III], equivalently the pushout along F of

$$0 \longrightarrow H_1(\mathcal{T}^\dagger;\mathbb{Q}) \longrightarrow C_1(\mathcal{T}^\dagger;\mathbb{Q}) \longrightarrow C_0(\mathcal{T}^\dagger;\mathbb{Q}) \longrightarrow \mathbb{Q} \longrightarrow 0,$$

where C_i is the rational cellular chain group. If a subgroup fixes a good vertex w_0, choose a path s from v to w_0 and take $\tau(g) = s - gs$. The cycle in (4.2) is then zero. Hence a_F vanishes on every good-vertex stabilizer.

Consider the bad line $\mathbb{Q}e_1$. Write P_L for its stabilizer in $K^\dagger$, $U = \langle U_{12}, U_{13}\rangle \cong \mathbb{Z}^2$ for its radical, and $L = P_L/U$ for its Levi quotient. The generators act on E_n by $e^{-Y\partial_X}$ and $e^{-Z\partial_X}$. A one-cocycle b with values b_1, b_2 therefore satisfies

$$(e^{-Y\partial_X} - I)b_2 = (e^{-Z\partial_X} - I)b_1.$$

Define its Bernoulli coordinate by

$$\alpha = B(-Y\partial_X)b_1, \qquad \beta = B(-Z\partial_X)b_2, \qquad \mathcal{J}(b) = Z\alpha - Y\beta. \tag{4.3}$$

The finite-difference identity gives $Y\partial_X\beta = Z\partial_X\alpha$, so $\partial_X \mathcal{J}(b) = 0$. This induces an isomorphism

$$H^1(U, E_n) \xrightarrow{\sim} \mathbb{Q}[Y,Z]_{n+1}.$$

Indeed, a coboundary has $\alpha = -Y\partial_X e$ and $\beta = -Z\partial_X e$, so its coordinate is zero. Conversely, $Z\alpha = Y\beta$ gives $\alpha = Yh$, $\beta = Zh$ because Y, Z are relatively prime. Choose $e \in E_n$ with $\partial_X e = -h$. Invertibility of the Bernoulli operators makes b its coboundary. For surjectivity, given $J \in \mathbb{Q}[Y,Z]_d$, $d = n+1$, take

$$b_1 = \alpha = \frac{J_Z}{d}, \qquad b_2 = \beta = -\frac{J_Y}{d}.$$

These are independent of X, hence satisfy the cocycle equation and are fixed by the Bernoulli operators. Euler's identity gives $\mathcal{J}(b) = J$.

For $1 \to U \to P_L \to L \to 1$, the *mixed* Hochschild–Serre term is $H^1(L, H^1(U, E_n))$, with one degree from each group. The coordinate above expresses its coefficients as binary polynomials. A binary cusp is an orbit of a rational line. A class is *cuspidal* when its restriction to every cusp stabilizer is zero. The next lemma identifies the mixed restriction of a_F and shows that its vanishing forces the literal polynomial J_f to vanish.

Lemma 4.1. *The restriction of a_F to the bad-line stabilizer has only the mixed Hochschild–Serre component. This component is cuspidal on the binary Levi. If it is zero, then $J_f = 0$ as a polynomial. In the coordinate* (4.3)*, its cocycle is the negative of the one whose specified order three generator has value J_f.*

Proof. The radical coordinate. Choose the base vertex $v = G_0$ and the path $s : G_0 \to \mathbb{Q}e_3 \to H_0$, where $G_0 = \langle e_1, e_3\rangle$ and $H_0 = \langle e_2, e_3\rangle$. The bad-line Levi fixes H_0 and its radical U fixes G_0. For Levi elements l, use $\tau(l) = s - ls$ and $\tau(ul) = u\tau(l)$. In particular $\tau(u)$ is zero. A binary Levi matrix l acts on the last two coordinates through $\mathrm{diag}(1, l)$. Put

$$A_0 = \begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}, \quad B_0 = \begin{pmatrix} -1 & 0 \\ -3 & 1 \end{pmatrix}, \quad R_0 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

When acting on $\mathbb{Q}^3$, these binary matrices are embedded as $\mathrm{diag}(1, g)$ in the bad-line Levi. Both B_0, R_0 fix s. The radical cocycle $\phi_l(u) = F((u-1)\tau(l))$ therefore has zero values for these two Levi elements. We make the path cancellation explicit. Put

$$a = e_3, \quad b = (1,1,2), \quad c = (0,1,2), \quad d = (1,0,1), \quad e = (2,1,2), \quad G' = \langle e_1, c\rangle.$$

Vectors denote their rational lines. Since $A_0 e_3 = -c$ and $A_0 H_0 = H_0$, the path $\tau(A_0) = s - A_0 s$ is

$$G_0 \longrightarrow a \longrightarrow H_0 \longrightarrow c \longrightarrow G'.$$

For independent vectors v, w, write $\mathfrak{e}(v, w) = [\mathbb{Q}v, \langle v, w\rangle] + [\langle v, w\rangle, \mathbb{Q}w]$. The apartment convention is $\mathcal{A}(v, w, t) = \mathfrak{e}(v, w) + \mathfrak{e}(w, t) + \mathfrak{e}(t, v)$. Cancelling repeated edges in the two translated paths gives

$$(U_{12} - 1)\tau(A_0) = \mathfrak{e}(a, b) + \mathfrak{e}(b, c) + \mathfrak{e}(c, a),$$
$$(U_{13} - 1)\tau(A_0) = \mathfrak{e}(a, d) + \mathfrak{e}(d, e) + \mathfrak{e}(e, c) + \mathfrak{e}(c, a).$$

The lines b, c, e lie in one plane, so $\mathfrak{e}(e, b) + \mathfrak{e}(b, c) = \mathfrak{e}(e, c)$. Consequently the two translated paths reduce to apartments:

$$(U_{12} - 1)\tau(A_0) = \mathcal{A}(e_3, (1,1,2), (0,1,2)),$$
$$(U_{13} - 1)\tau(A_0) = \mathcal{A}(e_3, (1,1,2), (0,1,2)) + \mathcal{A}(e_3, (1,0,1), (2,1,2)) + \mathcal{A}(e_3, (2,1,2), (1,1,2)).$$

The column-sign lists are $(1, -1, -1), (1, 1, -1), (1, -1, -1)$. By (4.1), the preceding apartment identities give

$$\phi_{A_0}(U_{12}) = \psi_1, \qquad \phi_{A_0}(U_{13}) = \psi_2,$$

where ψ_1, ψ_2 are exactly the symmetric frame polynomials in (2.7). Applying the radical coordinate (4.3), we obtain

$$\mathcal{J}(\phi_{A_0}) = ZB(-Y\partial_X)\psi_1 - YB(-Z\partial_X)\psi_2 = J_f.$$

Thus J_f is the literal radical coordinate and is independent of X. We write it as $J_f(Y, Z)$ for the rest of this proof.

The mixed bar edge of a two-cocycle is $e(z)(l)(u) = z(l, l^{-1}ul) - z(u, l)$. Our choices $\tau(u) = 0$ and $\tau(ul) = u\tau(l)$ give

$$z_F(l, l^{-1}ul) = F(\tau(l) - u\tau(l)) = -\phi_l(u) \qquad \text{and} \qquad z_F(u, l) \;\; = F(u\tau(l) - \tau(ul)) = 0.$$

Thus the mixed edge is $-\phi_l(u)$ and $z_F|_{U\times U} = 0$. We retain this sign, though its reversal would not affect vanishing. The Levi has rational cohomological dimension one, so no further filtration term occurs in this kernel: the complete restriction lies in $H^1(L, H^1(U, E_n))$. Under $\mathrm{diag}(t, L)$ the coordinate (4.3) transforms by $J(v) \mapsto t^{-2}J(L^{-1}v)$. Indeed inverse conjugation on the radical contributes $t^{-1}L$, the coefficient action contributes $t^{-1}\det(L)^{-1}$, and the contraction $(Z, -Y)$ contributes $\det L$. In particular the independent integral line sign acts trivially.

The binary quotient and its cusps.

At each cusp we take translation coinvariants, then reflection invariants.

Write C_m for the cyclic group of order m and $\mathfrak{S}_3$ for the permutation group on three letters. The remaining arithmetic group is $\Gamma_2 = \{g \in \mathrm{GL}_2(\mathbb{Z}) : (0,1)g = (0,1) \pmod 3\}$, and it has a presentation

$$\Gamma_2 = \langle A_0, B_0\rangle * \langle R_0\rangle \cong \mathfrak{S}_3 * C_2. \tag{4.4}$$

To verify completeness, the determinant one subgroup maps injectively to an index four subgroup of $\mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm I_2\} \cong C_2 * C_3$. On its four cosets the two generators act with cycle types $(2,2)$ and $(3,1)$. The Bass–Serre quotient has four edges and four vertices, one C_3 vertex stabilizer, and all other stabilizers trivial. Schreier reduction uses the standard matrices $\mathsf{S} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\mathsf{T} = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. The representatives $1, \mathsf{S}, \mathsf{ST}, \mathsf{ST}^2$ give $\langle A_0, U_0 \mid A_0^3 = 1\rangle$, where $U_0 = \left(\begin{smallmatrix} 1 & 0 \\ 3 & 1 \end{smallmatrix}\right)$ and $A_0 = U_0\mathsf{T}^{-1}$. Matrix multiplication gives $R_0U_0R_0 = U_0^{-1}$ and $R_0A_0R_0 = U_0^{-1}A_0^{-1}U_0$. Set $B_0 = U_0R_0$ to obtain (4.4), with no extra mixed relation.

Transport to $(a,b) = (-Y, Z-2Y)$ and write $\widehat{J}(a,b) = J_f(-a, b-2a)$. The actions become $\mathcal{A}(a,b) = (-b, a-b)$, $\mathcal{B}(a,b) = (-a, b-a)$, and $\mathcal{R}(a,b) = (-a, b-4a)$. The positive cocycle $\mathcal{J}(\phi_l)$ has values $\widehat{J}, 0, 0$ on A_0, B_0, R_0. Their finite group relations give $(1+\mathcal{A}+\mathcal{A}^2)\widehat{J} = 0$ and $\widehat{J}(a,b) = -\widehat{J}(b,a)$. For $u = (2\widehat{J} + \mathcal{A}\widehat{J})/3$, one has $(1-\mathcal{A})u = \widehat{J}$ and $\mathcal{B}u = u$. Thus in $V = \mathbb{Q}[a,b]_{n+1}$ the actual mixed class has quotient coordinate

$$j_F = -[u] \in V/(V^{\langle \mathcal{A},\mathcal{B}\rangle} + V^{\mathcal{R}}). \tag{4.5}$$

The minus sign comes from the mixed bar edge above.

Let $\Gamma_1(3) = \Gamma_2 \cap \mathrm{SL}_2(\mathbb{Z})$. A binary cusp is an orbit of a rational line. There are exactly two cusp orbits, represented by the coordinate lines, with widths one and three. Indeed a primitive column $(p,q)^t$ with $3 \mid q$ can be signed and completed to a matrix in $\Gamma_1(3)$ as its first column. If $3 \nmid q$, it can be so completed as its second column, adjusting a Bezout complement by a multiple of that column. These two cases are invariant and exhaustive. Write $d = n+1$. At the lower cusp the translation is $Z \mapsto Z - 3Y$, whose degree d coinvariant is the Z^d coefficient. Its reflection fixes that coefficient and reverses the translation, so its full cusp H^1 vanishes. At the upper cusp the translation is $Y \mapsto Y - Z$, with coinvariant the Y^d coefficient. The reflection acts on H^1 by $(-1)^{d+1}$. Thus this restriction is automatically zero for even d.

For odd d, Lemma 2.2 gives $w \mid f$. All three frames have sum $Z - 3Y$, so $J_f(Y, 3Y) = 0$, or $\widehat{J}(a,-a) = 0$. For the upper translation $\mathcal{A}^2\mathcal{B}\mathcal{R}$, its cocycle value is $\widehat{J} + \mathcal{A}\widehat{J}$. On $Z = 0$, that is $b = 2a$, the cocycle relations give $(\widehat{J} + \mathcal{A}\widehat{J})(a, 2a) = -\widehat{J}(a,-a) = 0$. Both cusp restrictions therefore vanish in every parity.

Detection of the polynomial. Finally suppose $j_F = 0$. Write the positive cocycle as $(1-l)m$. Its zero values at B_0, R_0 make m invariant under their product U_0, hence $m = cY^d$. For odd d the reflection makes $c = 0$. For even d, its A_0 value is $J_f(Y,Z) = c(Y^d - (-2Y+Z)^d)$. But $J_f(0,Z) = Zf(Z,-X,X)$ vanishes by the opposite pair vanishing. Thus $c = 0$ here as well. This proves the detection assertion. □

4.2. **The symmetric obstruction vanishes.** We apply the boundary comparison of Proposition C.5, proved in Appendix C, to the connecting class a_F.

Proposition 4.2. *Every symmetric homogeneous solution satisfies $J_f = 0$.*

Proof. The degrees at most three are zero by Lemma 2.1. For $n \geq 4$, let a_F be the connecting class, and complexify its coefficient system. Vanishing after complexification implies rational vanishing. The class restricts to zero at every good vertex. Its bad-line restriction is mixed and cuspidal by Lemma 4.1. Its restriction at each flag is therefore zero: a flag contains either a good line, where the whole restriction vanishes, or a bad line, where it is a binary cusp restriction of that mixed class.

Let P_0 be the stabilizer of the bad plane $\langle e_1, e_2\rangle$ in $K^\dagger$. Its radical is $N_P = \langle U_{13}, U_{23}\rangle$, and its Levi quotient is $L_P \cong \mathrm{GL}_2(\mathbb{Z})$. The same gradient calculation used in the support proof gives

$$H^1(N_P, E_n) = \overset{-1}{\det}\mathrm{Sym}^{n+1}((\mathbb{Q}^2)^*), \qquad H^2(N_P, E_n) = \overset{-2}{\det}\mathrm{Sym}^{n}((\mathbb{Q}^2)^*).$$

The two opposite binary unipotents have no common invariant in positive degree, so the second module has no L_P-invariants. The Levi has rational cohomological dimension one, leaving only the mixed restriction.

At a binary cusp the mixed edge map is injective for the same dimension reason. Flag vanishing therefore makes the plane class cuspidal.

Let ξ_P and ξ_Q be the collections of plane and line restrictions. Let T be the map taking plane cusp data to the line constant term of the convergent cusp form sum constructed in Appendix C. Proposition C.5 gives $\xi_Q = T\xi_P$. It remains to show that $\xi_P = 0$.

In even degree the bad-plane binary module has odd degree $n+1$, so its central $-I_2$ kills its cohomology. Since all good-plane restrictions vanish, $\xi_P = 0$ and hence $\xi_Q = 0$.

In odd degree the bad-plane datum is a level one cusp form, as explained in Appendix C.6. On each classical Hecke eigenform its image at the good-line cusp is the nonzero scalar (C.16), in the seed normalizations of (C.4). Thus the good-line map is injective on the bad-plane space, whereas the good-line restriction of a_F is zero. Again both ξ_P and ξ_Q vanish.

Lemma 4.1 now gives $J_f = 0$ as a polynomial, not merely modulo coboundaries. □

4.3. **Extending the binary data.** Return to $\ell = (1,1,1)$ and the determinant-twisted model of Lemma 3.2. The three signed-permutation orbits on $\mathcal{S}$ are distinguished by support size. Thus extension requires two seeds besides f. Corollary 2.4 will supply them. We first allow rational functions to isolate the possible odd degree pole.

Lemma 4.3. *A signed-permutation equivariant function F as in* (3.3) *is uniquely determined by its seeds $P = F(1,0,0)$, $Q = F(1,1,0)$, and $f = F(1,1,1)$. Such an F satisfies* (3.4) *if and only if the seeds satisfy*

$$f \text{ symmetric}, \quad Q(a,b,c) = Q(b,a,c) = -Q(a,b,-c),$$
$$P(a,b,c) = P(a,c,b) = -P(a,-b,c) = -P(a,b,-c),$$
$$N_{12}f = N_{12}Q = N_{23}P = 0,$$

and

$$Q(a,b,c+a) - Q(a,b,c) = f(a+c,-c,b), \tag{4.6}$$
$$P(a,c+a,b) - P(a,c,b) = Q(-c,a+c,b). \tag{4.7}$$

Conversely, seeds with these properties extend uniquely by (3.3) *to a function satisfying both fiber equations. The statement holds for polynomial or rational function coefficients.*

Proof. The signed-permutation orbits have support sizes one, two, and three. Their stabilizers give the seed symmetries. The norm equations at

$$(0,0,1), \quad (1,1,0), \quad (1,1,1), \quad (1,0,1), \quad (1,0,0)$$

give the five displayed equations. To see that they exhaust the fibers, write $R(r)$ for the left side of (3.4), and put $D_{12} = \mathrm{diag}(-1,-1,1)$ and $D_3 = \mathrm{diag}(1,1,-1)$. Direct reindexing gives

$$R(rA) = R(r) \mid A, \qquad R(rD) = \chi_1(D)R(r) \mid D \quad (D = D_{12}, D_3).$$

The five representative orbits under $\langle A, D_{12}, D_3\rangle$ have sizes $2,2,4,12,6$, respectively. They partition the 26 covectors. □

For the finite expansion $f = \sum_j w^{e_j}\Phi(q_j)$ in (2.13), define the candidate seeds

$$\begin{aligned} Q(a,b,c) &= \sum_j (a+b)^{e_j}\big(q_j(a,c) + q_j(b,c)\big), \\ P(a,b,c) &= \sum_j a^{e_j} q_j(b,c). \end{aligned} \tag{4.8}$$

In even degree both seeds are polynomial. In odd degree only $e_0 = -1$ is negative. Oddness of q_0 in its first variable makes $q_0(a,c) + q_0(b,c)$ divisible by $a+b$, so Q is still polynomial. Only P can have a pole, with term $q_0(b,c)/a$. We remove it in Section 4.4.

The q_j give the required seed symmetries, and (2.4) gives (4.6)–(4.7) term by term. Also $N_2 q_j = 0$ gives $N_{23}P = 0$. Since $N_{12}f = 0$, only $N_{12}Q = 0$ remains.

Lemma 4.4. *Let $f \in \mathbb{Q}[x,y,z]$ be symmetric with $Nf = 0$. If $Q \in \mathbb{Q}[a,b,c]$ is symmetric in a, b, odd in c, and satisfies* (4.6)*, then $N_{12}Q = 0$.*

Proof. Let Δ_s translate c by s and subtract the original value. Put

$$R_1 = f(a+c,-c,b), \quad R_2 = f(c-b,-c,a-b), \quad R_3 = f(b-a+c,-c,-a).$$

The norm and symmetry give

$$\Delta_{-b}\Delta_{b-a}R_1 + \Delta_a\Delta_{b-a}R_2 + \Delta_a\Delta_{-b}R_3 = 0. \tag{4.9}$$

Appendix B.1 gives the twelve-norm expansion of this identity. By (4.6), it becomes

$$\Delta_a\Delta_{-b}\Delta_{b-a}(N_{12}Q) = 0.$$

Over $\mathbb{Q}(a,b)$, each difference equals its nonzero step times ∂_c times an invertible finite power series in ∂_c. Hence $\partial_c^3 N_{12}Q = 0$, and oddness leaves $N_{12}Q = c\kappa(a,b)$.

Evaluate at $c = a$. With $g_1 = f(a,0,b)$, $g_2 = f(-b,0,a-b)$, $g_3 = f(b-a,0,-a)$, and $h = f(a-b,-a,a-b)$, the mixed equation and symmetries give the three values $g_1, g_2 - h, -g_3$. Since $(Nf)(a,b,0) = g_1 + g_2 + g_3$ and $(Nf)(b-a,0,-a) = 2g_3 + h$, we get

$$a\kappa = g_1 + g_2 - h - g_3 = (g_1 + g_2 + g_3) - (2g_3 + h) = 0.$$

Thus, we have $\kappa = 0$. □

In even degree, Lemmas 4.3 and 4.4 give a polynomial extension with identity apartment value f, so Proposition 3.4 gives $f \in U_n$. The odd degree case remains polynomial except for the possible term $q_0(b,c)/a$.

4.4. **A pole detected by Hecke operators.** To eliminate $q_0(b,c)$ using Hecke operators, we enlarge the coefficient module to allow simple poles along every rational hyperplane. This makes it stable under rational changes of variables.

Define

$$\mathcal{M}_n = V_n + \sum_{0 \neq L \in (\mathbb{Q}^3)^*} \frac{V_{n+1}}{L}.$$

The sums are finite, and each p/L has homogeneous degree n. This is a vector space enlargement, not a localization allowing products of linear denominators.

For a rational hyperplane $H = \{L = 0\}$, define

$$\mathcal{R}_{n,H} = (V_{n+1}/LV_n) \otimes_{\mathbb{Q}} (\mathbb{Q}L)^*, \qquad \mathcal{R}_n = \bigoplus_H \mathcal{R}_{n,H}.$$

The quotient records the numerator on H, and the dual factor removes the choice of its equation. With $L^\vee(L) = 1$, set

$$\operatorname{res}_H(p/L) = [p] \otimes L^\vee.$$

Replacing (p, L) by (cp, cL) multiplies the factors by c, c^{-1}, so the residue is unchanged. The direct sum over rational hyperplanes through the origin has finite support.

Taking residues gives a $\mathrm{GL}_3(\mathbb{Q})$-equivariant exact sequence

$$0 \longrightarrow V_n \longrightarrow \mathcal{M}_n \xrightarrow{\text{res}} \mathcal{R}_n \longrightarrow 0. \tag{4.10}$$

At a generic point of a hyperplane, terms with other pole hyperplanes are regular, so residues are independent. Choosing numerators lifts every residue. After combining proportional denominators, zero residue means the numerator is divisible by its denominator, proving exactness.

This surjectivity is on coefficients, not on homology. The next lemma identifies top residue classes with periods and constrains those that lift. Write $[ac^n]q$ for the coefficient of ac^n.

Lemma 4.5. *The map sending $q \in \mathcal{B}_{n+1}$ to the residue cycle represented by*

$$F_q(a,b,c) = \frac{q(b,c)}{a} + \frac{q(a,c)}{b} + \frac{q(a,b)}{c}$$

identifies $\mathcal{B}_{n+1}$ with $H_3(G, \mathcal{R}_n \otimes \det)$. Its inverse reads the literal residue at $a = 0$, namely $q(b,c)$. Let $\mathcal{L}_n$ be the image of $H_3(G, \mathcal{M}_n \otimes \det)$ in this space. If $n > 0$, then $[ac^n]q(a,c) = 0$ for every $q \in \mathcal{L}_n$. Moreover $\mathcal{L}_n$ is stable under the usual binary Hecke operators.

Proof. A residue cycle is symmetric and separately odd, with finite hyperplane support. Choose a noncoordinate supported primitive normal of maximal height $|v_1|+|v_2|+|v_3|$, and normalize it by signed permutation to $v = (\alpha, \beta, \gamma)$ with $\alpha, \beta > 0$, $\gamma \geq 0$. The normals $vA = (\beta, -\alpha - \beta, \gamma)$ and $vA^2 = (-\alpha - \beta, \alpha, \gamma)$ remain primitive and have larger height, so neither is supported. The norm relation at the hyperplane normal to v would then contain only its nonzero residue, a contradiction. Thus only coordinate poles occur. Symmetry gives the representative F_q in the statement, with q symmetric and separately odd. Writing $d = a - b$, direct substitution gives

$$N_{12}F_q = K_q + \frac{N_2 q(a,b)}{c}, \qquad K_q = \frac{a^2 q(a,c) - b^2 q(b,c) - d^2 q(d,c)}{abd}. \tag{4.11}$$

The numerator vanishes at $a = 0$, $b = 0$, and $a = b$. The three linear factors $a, b, a - b$ are pairwise relatively prime, so their product divides the numerator and K_q is polynomial. Consequently F_q is norm-zero modulo polynomial coefficients if and only if $N_2 q = 0$. Together with its symmetry and separate oddness, this proves the converse as well as the residue identification.

It remains to determine the liftable residue subspace $\mathcal{L}_n$. If this residue lifts to a top cycle, write the lift as $F_q + P$. The polynomial remainder P is symmetric and separately odd, with $NP = -K_q$. To constrain q, put $\lambda = [ac^n]q(a,c)$ and compare the c^n coefficients. The numerator of K_q contributes

$$\lambda\{a^3 - b^3 - (a-b)^3\} = 3\lambda ab(a-b),$$

so $[c^n]K_q = 3\lambda$. Every monomial of P is divisible by ab, and P has total degree n. Its c-degree is therefore at most $n - 2$. The norm acts only on a, b, so the same bound holds for NP. Comparing the c^n coefficient gives $3\lambda = 0$ and therefore $\lambda = 0$.

A Hecke correspondence consists of restriction, rational matrix transport, and transfer. Sequence (4.10) commutes with all three: rational matrices preserve numerator degree and pole order while permuting hyperplanes, and the other maps are natural in the coefficients. Thus liftable residues are Hecke-stable. Let $T_3(p,1)$ be the double-coset operator of $\mathrm{diag}(1,1,p)$ and $T_2(p)$ the usual binary prime operator. In the same literal residue coordinate, these operators satisfy

$$T_3(p,1) = pT_2(p) + p^2 I. \tag{4.12}$$

Lemma B.1 identifies the literal residue with the binary period coordinate. Appendix B.2 then computes the transfer using Hecke-equivariant Shapiro [5, Theorem 3.1]. Naturality makes $\mathcal{L}_n$ stable under the left side of (4.12). Subtracting $p^2 I$ and dividing by p gives binary stability. Only the nonzero coefficient of $T_2(p)$ is needed for stability. A fixed scalar in the period comparison does not change this relation. □

Put $d = n + 1$. By Lemma 4.5, $\mathcal{L}_n \subseteq \mathcal{B}_d$ is Hecke-stable and $[XY^{d-1}]q(X,Y) = 0$ for every $q \in \mathcal{L}_n$. The coefficient condition alone does not imply vanishing. Hecke stability is essential.

Suppose that $\mathcal{L}_n \otimes_{\mathbb{Q}} \mathbb{C}$ is nonzero. The odd Eichler–Shimura correspondence identifies the Hecke action on $\mathcal{B}_d \otimes_{\mathbb{Q}} \mathbb{C}$ with the usual action on level one cusp forms of weight $k = d + 2$ [17, Theorem 2.1 and Section 5]. Since the Hecke operators commute and the space is finite dimensional, $\mathcal{L}_n \otimes_{\mathbb{Q}} \mathbb{C}$ contains a nonzero simultaneous eigenperiod q_h, corresponding to a normalized cusp eigenform h of weight k. Rescaling q_h within this subspace, we may take it to be the odd part in X of the period polynomial (1.8). Its indicated coefficient is then

$$[XY^{d-1}]q_h = -d \int_0^{i\infty} h(z) z^{d-1}\, dz = -d\, i^d \frac{\Gamma(d)}{(2\pi)^d} L(h,d). \tag{4.13}$$

A nonzero level one cusp form has weight at least twelve. Since $d = k - 2$, Lemma 2.5 gives $L(h,d) \neq 0$. The coefficient in (4.13) is therefore nonzero, contradicting its vanishing on $\mathcal{L}_n \otimes_{\mathbb{Q}} \mathbb{C}$. Thus, we have $\mathcal{L}_n = 0$.

For the odd degree expansion, the three seeds define a rational 26-coordinate cycle F. Set

$$S = \sum_{r \in \mathcal{S}} F(r) \in \mathcal{M}_n.$$

Summing the signed-permutation relations shows that S is symmetric and odd in each variable. Summing the fiber norm equations and reindexing by $r \mapsto rA$ gives $NS = 0$. Thus S represents a top cycle for the full group G with coefficients in $\mathcal{M}_n \otimes \det$. By Lemma 4.5, its residue corresponds to an element of $\mathcal{L}_n = 0$. In particular, its literal residue along $a = 0$ vanishes.

Only $F(1,0,0)$ and $F(-1,0,0)$ contribute to the residue at $a = 0$: Q, f are polynomial, and the other translates of P have poles only at $b = 0$ or $c = 0$. Both contributing coordinates equal P, since

$$F(-1,0,0) = -P(-a,b,c) = P(a,b,c),$$

because every exponent $e_j = 2j - 1$ in (4.8) is odd. Consequently, we have

$$0 = \operatorname{res}_{a=0} S = 2q_0(b,c),$$

and hence $q_0 = 0$. The possible pole in P disappears, so all three seeds are polynomial. Proposition 3.4 applies to the resulting ordinary extension and gives $f \in U_n$.

We have proved symmetric support in both parities. Since $n > 0$, we have

$$0 = (Nf)(0,0,z) = 3f(0,0,z).$$

Symmetry therefore eliminates all pure powers. Every remaining monomial involves exactly two variables, so

$$f = \Phi(q), \qquad q(X,Y) = f(X,Y,0).$$

The polynomial q is symmetric and vanishes on both axes.

Restricting $Nf = 0$ to $z = 0$ gives $N_2 q = 0$. Evaluating this binary relation at (X,X) and using the axis conditions gives $q(X,X) = 0$. We may now evaluate the full norm at $(x,0,z)$:

$$0 = (Nf)(x,0,z) = 2\{q(x,z) + q(-x,z)\}.$$

Thus q is odd in its first variable, and symmetry makes it odd in the second as well.

If n is odd, separate oddness forces $q = 0$, and hence $f = 0$. If n is even, these are precisely the conditions on q in Theorem 1.2. Conversely, let q satisfy those conditions. In the expansion of $N\Phi(q)$, the three terms involving only x, y sum to $N_2 q(x,y) = 0$, while the other six cancel in pairs by symmetry and separate oddness. Thus $\Phi(q)$ is a symmetric solution, and its restriction to $z = 0$ is q. This completes the symmetric classification.

5. Alternating solutions

In even degree we construct the extension and apply Proposition 3.4. In odd degree a separate mod four argument proves vanishing, without the Bernoulli obstruction or arithmetic cohomology.

5.1. **The even degree extension.** The omitted vertex and edge stabilizers control the extension obstruction.

Proposition 5.1. *Every alternating even degree solution belongs to U_n.*

Proof. Vertex stabilizers. Assume $n > 0$, since the constant case is zero. For the stabilizer calculation use the group $K^\dagger$ and graph $\mathcal{T}^\dagger$ introduced in Section 4. The relative graph has two vertex orbits, represented by the bad line $\mathbb{Q}e_1$ and bad plane $\mathbb{Q}e_1 + \mathbb{Q}e_2$, and two edge orbits: bad line in bad plane and bad line in a good plane. Integral parabolic reduction modulo three proves this list exhaustive. The vertex stabilizers are

$$H_P = \left\{ \begin{pmatrix} A & u \\ 0 & 1 \end{pmatrix} \right\}, \qquad H_L = \left\{ \begin{pmatrix} \eta & u \\ 0 & D \end{pmatrix} \right\},$$

where $A \in \mathrm{GL}_2(\mathbb{Z})$, $u \in \mathbb{Z}^2$, $\eta = \pm 1$, and $D \in \Gamma_2 = \{D \in \mathrm{GL}_2(\mathbb{Z}) : (0,1)D = (0,1) \pmod 3\}$. The edge stabilizers are

$$H_{BB} = \left\{ \begin{pmatrix} \eta_1 & a & b \\ 0 & \eta_2 & c \\ 0 & 0 & 1 \end{pmatrix} \right\}, \qquad H_{BG} = \left\{ \begin{pmatrix} \eta_1 & a & b \\ 0 & \eta_2 & 0 \\ 0 & 3k & 1 \end{pmatrix} \right\},$$

with integral entries and independent signs η_i.

Take $E = \mathrm{Sym}^n((\mathbb{Q}^3)^*)$ and write ρ for its left inverse substitution representation. The vertex radicals are $N_P = \langle U_{13}, U_{23} \rangle$ and $N_L = \langle U_{12}, U_{13} \rangle$. On dual polynomial coefficients, the generators act with $\log \rho(U_{ij}) = -x_j \partial_{x_i}$. For N_P, the closed pairs are gradients of potentials, as in Section 3. Modulo exact pairs, restriction of the potential to $z = 0$ gives a binary form of degree $n+1$. Under a general diagonal matrix this identification has an extra factor d_3, which is one on H_P. For N_L, the contraction $(\alpha, \beta) \mapsto z\alpha - y\beta$ gives the binary form. Inverse conjugation on its two roots makes this contraction transform by $\eta^{-1} \det D$ times the dual binary action. In the table, $\mathrm{Sym}^r(x,y)^*$ abbreviates the degree r binary polynomials with inverse substitution action, and similarly for y, z. These calculations give its first column.

	$H^1(N, E)$	$H^2(N, E)$
N_P	$\mathrm{Sym}^{n+1}(x,y)^*$	$\det(A)^{-1} \mathrm{Sym}^n(x,y)^*$
N_L	$\eta^{-1} \det(D) \mathrm{Sym}^{n+1}(y,z)^*$	$\eta^{-n-2} \det(D)$

TABLE 1. Cohomology of the two vertex radicals with $E = \mathrm{Sym}^n((\mathbb{Q}^3)^*)$ and Levi actions.

The H^2 column follows from the top cohomology formula for a rank two lattice, $H^2(N, E) = E_N \otimes \bigwedge^2 (N \otimes \mathbb{Q})^*$, using $E_{N_P} = \mathbb{Q}[x,y]_n$ and $E_{N_L} = \mathbb{Q}x^n$. These are coinvariants. The corresponding invariant spaces are $E^{N_P} = \mathbb{Q}z^n$ and $E^{N_L} = \mathbb{Q}[y,z]_n$. These invariant spaces enter the edge calculation below. Both Levis are virtually free. Thus total degree two in Hochschild–Serre has only $H^1(L, H^1(N, E))$ and $H^0(L, H^2(N, E))$. At H_P, the central $-I_2$ acts by -1 on H^1 and positive binary degree kills the H^2 invariants. At H_L, the independent line sign kills H^1 and the reflection $\mathrm{diag}(-1,1) \in \Gamma_2$ kills H^2. Hence, we have $H^2(H_P, E) = H^2(H_L, E) = 0$.

Edge stabilizers. For either edge radical, extend a rank two radical by its remaining cyclic root. The exact sequence $0 \to H^1(\mathbb{Z}, E^N) \to H^1(N_{\mathrm{edge}}, E) \to H^1(N, E)^{\mathbb{Z}} \to 0$ is equivariant for $D_0 = \{\mathrm{diag}(\eta_1, \eta_2, 1)\}$. A weight (a_1, a_2, a_3) means the character $\mathrm{diag}(d_1, d_2, d_3) \mapsto d_1^{a_1} d_2^{a_2} d_3^{a_3}$. Table 2 records the weights of the two one-dimensional subquotients:

edge type	first subquotient	second subquotient
BB	$(-1, 1, -n)$	$(0, -n-1, 1)$
BG	$(0, 1, -n-1)$	$(-1, -n, 1)$

TABLE 2. Weights of the two degree-one cohomology subquotients for each edge stabilizer. For even n, each row has a nontrivial sign character.

For type BB, use N_P and the remaining root E_{12}. The left term of the five-term sequence comes from $E^{N_P} = \mathbb{Q}z^n$ and the dual of E_{12}, giving $(-1, 1, -n)$. The invariant potential in the right term is y^{n+1}, with the extra d_3 factor, giving $(0, -n-1, 1)$. For type BG, use N_L and E_{32}. The coinvariant of $E^{N_L} = \mathbb{Q}[y,z]_n$ under $-y\partial_z$ is represented by z^n. Including the root-dual factor gives $(0, 1, -n-1)$. The right term is represented by y^{n+1} in $\eta^{-1} \det(D) \mathrm{Sym}^{n+1}(y,z)^*$, giving $(-1, -n, 1)$. For H_{BG} the actual remaining lattice root is $3E_{32}$. Multiplication of its logarithmic differential by the nonzero rational scalar 3 changes neither its kernel nor its cokernel. Since n is even, each weight has a nontrivial first or second sign. Exact averaging gives $H^1(H_{BB}, E) = H^1(H_{BG}, E) = 0$.

Extension to the full graph. Put $\mathrm{St}^\dagger = H_1(\mathcal{T}^\dagger;\mathbb{Q})$ and $Q_0 = \mathrm{St}\,/\,\mathrm{St}^\dagger$. The relative chain sequence and connectedness of the two graphs give

$$0 \longrightarrow Q_0 \longrightarrow \mathbb{Q}[K^\dagger/H_{BB}] \oplus \mathbb{Q}[K^\dagger/H_{BG}] \longrightarrow \mathbb{Q}[K^\dagger/H_L] \oplus \mathbb{Q}[K^\dagger/H_P] \longrightarrow 0.$$

Applying $\mathrm{Hom}_{K^\dagger}(-,E)$ and Shapiro gives the exact segment

$$H^1(H_{BB},E) \oplus H^1(H_{BG},E) \longrightarrow \mathrm{Ext}^1_{K^\dagger}(Q_0,E) \longrightarrow H^2(H_L,E) \oplus H^2(H_P,E).$$

The four vanishings make $\mathrm{Ext}^1_{K^\dagger}(Q_0,E) = 0$. The sequence $0 \to \mathrm{St}^\dagger \to \mathrm{St} \to Q_0 \to 0$ shows that $\mathrm{Hom}_{K^\dagger}(\mathrm{St},E) \to \mathrm{Hom}_{K^\dagger}(\mathrm{St}^\dagger,E)$ is surjective. Returning to the original coordinates, we apply Proposition 3.4 on the identity apartment. This value is the original polynomial, so it belongs to U_n. □

5.2. **Odd degree and reduction modulo four.** Recall that $N = I + A + A^2$, with A defined in (2.1). The following integral statement implies that every homogeneous alternating solution of odd degree vanishes over a field of characteristic zero.

Theorem 5.2. *Let $f \in \mathbb{Z}[x,y,z]$ be alternating, with every monomial of odd total degree. If $Nf \in 4\mathbb{Z}[x,y,z]$, then $f \in 2\mathbb{Z}[x,y,z]$. Consequently, an alternating homogeneous solution of $Nf = 0$ in odd degree over any characteristic zero field is zero.*

The integral assertion is stronger than the characteristic zero vanishing. It is also sharp with respect to the modulus: if $f = (x+y+z)^2(x-y)(x-z)(y-z)$, direct expansion gives

$$Nf = 6xy(x-y)(x^2-xy+y^2-z^2).$$

Thus divisibility of Nf by two does not force f to have even coefficients.

The proof uses an explicit reduced norm-kernel calculation. We prove the kernel description in Appendix D. The finite identities are identities of polynomials, not experimental checks in a range of degrees.

5.3. **The eight coefficient polynomials.** Alternation becomes symmetry over $k = \mathbb{F}_2$, so we need the symmetric kernel of the reduced norm

$$\overline{N}g = g(x,y,z) + g(y,x+y,z) + g(x+y,x,z).$$

Put

$$a = x+y+z, \qquad b = xy+xz+yz, \qquad c = xyz, \qquad C_+ = ab+c.$$

The symmetric ring is $k[a,b,c]$. Give its elementary symmetric generators weights $1,2,3$, respectively, so weighted degree agrees with total degree in x,y,z.

Define

$$d_4 = a^4 + b^2 + ac, \qquad d_6 = a^2b^2 + a^3c + abc + c^2, \qquad d_7 = ac(ab+c). \tag{5.1}$$

These are the Dickson invariants of degrees four, six, and seven [13]. Their invariance under invertible linear substitutions over k follows from the product identity

$$\prod_{(v_1,v_2,v_3)\in k^3} (T + v_1x + v_2y + v_3z) = T^8 + d_4T^4 + d_6T^2 + d_7T. \tag{5.2}$$

An invertible substitution permutes the eight factors, proving invariance. Multiplication by these invariants therefore preserves the symmetric kernel of $\overline{N}$.

Lemma 5.3. *The ring $R = k[d_4,d_6,d_7]$ is a polynomial ring. On symmetric polynomials, $\ker\overline{N}$ is a free R-module on $K_j = aC_+B_j$, $0 \le j < 8$, where*

$$\begin{aligned}(B_0,\dots,B_7) = (&1, a, a^2+b, ab, a^3+c, aC_+,\\ &a^3(a^2+b), a^6+a^3c+b^3+c^2).\end{aligned} \tag{5.3}$$

The proof and finite data are in Appendix D. Put

$$p = d_4, \qquad q = d_6, \qquad \rho = d_7^2, \qquad R_0 = k[p, q, \rho], \qquad R_2 = k[p^2, q^2, \rho].$$

The degrees of K_j are

$$(\delta_0, \ldots, \delta_7) = (4, 5, 6, 7, 7, 8, 9, 10).$$

Let $\kappa_j = 1$ when δ_j is even and $\kappa_j = 0$ otherwise, and set

$$L_j = K_j d_7^{\kappa_j}. \tag{5.4}$$

Since d_4, d_6 have even degree and d_7 has odd degree, Lemma 5.3 gives the following unique expression for every odd degree symmetric polynomial in the reduced norm kernel:

$$v = \sum_{j=0}^{7} r_j(p, q, \rho) L_j, \qquad r_j \in R_0. \tag{5.5}$$

This expression is valid in all degrees. It is the reason a finite calculation can prove Theorem 5.2 uniformly.

5.4. **An explicit mod four obstruction.** An *integral lift* of $v \in k[x, y, z]$ is a polynomial $\widetilde{v} \in \mathbb{Z}[x, y, z]$ reducing to v. If $\overline{N}v = 0$, its norm is even, but need not be zero. Divide the norm by two, reduce modulo two, and apply the following operator Π to obtain an obstruction independent of the lift within each symmetry type.

We write a linear substitution as a triple of linear forms: $(\lambda_1, \lambda_2, \lambda_3)$ sends (x, y, z) to $(\lambda_1, \lambda_2, \lambda_3)$. Let $\mathscr{T}$ consist of

$$\begin{gathered}
(y, z, x+y), \quad (z, y, x+z), \quad (x, z, x+y), \\
(x, z, y+z), \quad (x+y, z, x), \quad (x+y, z, y), \\
(z, x+y+z, x+z), \quad (x+y+z, z, y+z), \quad (x, y, x+z), \\
(x, y, y+z), \quad (y, x+z, x), \quad (y, x+y+z, x+y), \\
(x+z, y, z), \quad (x+y+z, y, y+z), \quad (x, y+z, y), \\
(x, x+y+z, x+y), \quad (x, y+z, z), \quad (x, x+y+z, x+z).
\end{gathered} \tag{5.6}$$

Each substitution is invertible over k. For $h \in k[x, y, z]$, define

$$\Pi h(x, y, z) = \sum_{(\lambda_1, \lambda_2, \lambda_3) \in \mathscr{T}} h(\lambda_1, \lambda_2, \lambda_3). \tag{5.7}$$

Lemma 5.4. *For every symmetric $g \in k[x, y, z]$, one has $\Pi \overline{N} g = 0$.*

Proof. Expand the left side into 54 terms. Identify triples of linear forms that differ only by a permutation, which does not change their value on a symmetric polynomial. There are exactly 27 resulting triples, each occurring twice. They cancel in characteristic two. The lists in (5.6) and the three substitutions defining $\overline{N}$ completely specify these pairings. □

For v as in (5.5), choose an alternating integral lift $\widetilde{v}^-$. Such a lift exists: in the displayed generators replace the factor aC_+ by

$$a\Delta, \qquad \Delta = (x - y)(x - z)(y - z),$$

and lift all remaining symmetric expressions integrally. Indeed, we have $\overline{\Delta} = C_+$ in characteristic two. Since $\overline{N}v = 0$, the coefficients of $N\widetilde{v}^-$ are even. Define

$$\beta_-(v) = \Pi \left(\frac{N\widetilde{v}^-}{2} \bmod 2 \right). \tag{5.8}$$

Define $\beta_+(v)$ in the same way with any symmetric integral lift.

Lemma 5.5. *The maps $\beta_\pm$ are independent of the respective choices of lift and are R_2-linear.*

Proof. Two lifts with the same symmetry differ by $2h$ with that symmetry. The reduction $\overline{h}$ is symmetric in either case, so the change in (5.8) is $\Pi\overline{Nh} = 0$ by Lemma 5.4.

If an integral polynomial D is invariant modulo two, its square is invariant modulo four: $(D+2E)^2 \equiv D^2 \pmod 4$. Consequently, the integral squares of the three polynomials in (5.1) are invariant modulo four under the substitutions in N. Their reductions are also invariant under every substitution in (5.6). Multiplying a lift by any polynomial in those squares therefore commutes with division of its norm by two followed by reduction and application of Π. These squares generate R_2. □

5.5. **The triangular calculation and the proof.** Write $v = \sum_{j=0}^{7} r_j(p,q,\rho)L_j$ as in (5.5). Subscripts p, q denote formal derivatives with ρ fixed. Define

$$\begin{aligned} U(v) = {} & r_{4,p}L_0 + \rho r_{5,p}L_1 + r_{4,q}L_2 \\ & + (r_4 + \rho r_{5,q})L_3 + \rho(r_{5,q} + r_{7,p})L_4 + \rho r_{7,q}L_6. \end{aligned} \tag{5.9}$$

Only r_4, r_5, r_7 enter U.

Lemma 5.6. *On the odd reduced norm kernel, we have*

$$\beta_+(v) = U(v), \qquad \beta_-(v) = v + U(v). \tag{5.10}$$

Moreover, we have $U^3 = 0$.

Proof. Splitting powers of p, q into even and odd parts gives the R_2-basis $1, q, p, pq$ of R_0. Lemma 5.5 therefore reduces (5.10) to the 32 polynomials

$$V_{4j} = L_j, \quad V_{4j+1} = qL_j, \quad V_{4j+2} = pL_j, \quad V_{4j+3} = pqL_j.$$

Their degrees are at most 27. Use the integral polynomial formulas above to form each norm, divide its even coefficients by two, and apply the 18 substitutions in (5.6). Exact expansion gives $\beta_-(V_\nu) = V_\nu + \beta_+(V_\nu)$ in every column. All nonzero β_+ columns are the following:

ν	$\beta_+(V_\nu)$	ν	$\beta_+(V_\nu)$
16	V_{12}	17	$V_8 + V_{13}$
18	$V_0 + V_{14}$	19	$V_1 + V_{10} + V_{15}$
21	$\rho(V_{12} + V_{16})$	22	ρV_4
23	$\rho(V_5 + V_{14} + V_{18})$	29	ρV_{24}
30	ρV_{16}	31	$\rho(V_{17} + V_{26})$

(5.11)

The other 22 columns are zero. This is exactly (5.9), since formal differentiation in p, q is R_2-linear. Thus the finite check proves the identity in every degree, not only through degree 27.

The image of U has no L_5 or L_7 coefficient. Applying U again therefore leaves only L_0, L_2, L_3 terms, and U kills those terms. Hence, we have $U^3 = 0$. □

Proof of Theorem 5.2. Reduce f modulo two, obtaining v. Alternation becomes symmetry and $Nf \equiv 0 \pmod 4$ gives $\overline{N}v = 0$. Using f itself as the lift in (5.8), we get $\beta_-(v) = 0$. By Lemma 5.6, this gives $(I+U)v = 0$. But $(I+U)^{-1} = I + U + U^2$, so we have $v = 0$.

A nonzero rational alternating solution of odd degree can be scaled to a primitive integer polynomial. The integral assertion would make all its coefficients even, a contradiction. In each fixed degree the norm and symmetry conditions are a finite rational linear system. Its kernel remains zero after extension to any characteristic zero field. □

6. Conclusion of the proof

We assemble the rational cases and extend coefficients. For a commutative $\mathbb{Q}$-algebra R, define $W_n(R)$ and $\mathscr{P}_n^0(R)$ by the same polynomial identities with coefficients in R.

Proposition 6.1. *For every commutative $\mathbb{Q}$-algebra R and every $n \geq 0$, the natural maps*

$$R \otimes_{\mathbb{Q}} W_n \xrightarrow{\sim} W_n(R) \qquad \text{and} \qquad R \otimes_{\mathbb{Q}} \mathscr{P}_n^0 \xrightarrow{\sim} \mathscr{P}_n^0(R) \tag{6.1}$$

are isomorphisms. The same statement holds for the symmetric and alternating solution spaces and for the even and odd period spaces. The ambient coefficient-extension maps commute with restriction to a coordinate plane and with cyclic summation.

Proof. In monomial bases, the defining identities are finite linear systems over $\mathbb{Q}$. Every $\mathbb{Q}$-algebra is flat over $\mathbb{Q}$, so extension of scalars preserves their kernels. This proves the two isomorphisms and applies unchanged after imposing symmetry or parity. Restriction to a coordinate plane and cyclic summation have rational matrices, so they commute with extension of scalars as well. □

Proof of Theorems 1.1 and 1.2. Lemma 2.1 reduces to four homogeneous symmetry-parity sectors. Section 4 proves the symmetric classification, Proposition 5.1 gives alternating even degree support, and Theorem 5.2 gives alternating odd degree vanishing. Thus every solution has two-variable support and every odd piece vanishes. The symmetric summand is $f(X, Y, 0)$, and its converse follows from the pairwise cancellation at the end of Section 4.

The operator $\partial_x\partial_y\partial_z$ kills exactly the monomials missing a variable. The remaining monomials have distinct nonzero derivatives in characteristic zero, proving the differential formulation.

Proposition 6.1 writes every solution over a characteristic zero field as a linear combination of rational solutions. It therefore preserves support and odd degree vanishing. Its compatibility with restriction preserves the symmetric classification and the unique choice $q = f(X, Y, 0)$. This proves both theorems. □

7. Period polynomials, dimensions, and the common divisor

We identify the alternating binary restrictions, prove the introductory corollaries, and describe the dependence on the coefficient ring, including the failure of support in positive characteristic.

7.1. **The complete binary restriction.** Theorem 1.2 gives the symmetric case. For alternating solutions, support and odd degree vanishing leave the following positive even degree classification.

Lemma 7.1. *Let F be a field of characteristic zero and let $n > 0$ be even. An alternating polynomial $f \in F[x, y, z]_n$ is a solution if and only if $f = \Phi(q)$, where $q \in F[X, Y]_n$ is alternating, even in each variable, zero on both axes, and satisfies* (1.5)*. In that case, we have $q(X, Y) = f(X, Y, 0)$.*

Proof. Suppose that f is a solution. Theorem 1.1 gives its two-variable support. The norm at $(0, 0, z)$ gives $3f(0, 0, z) = 0$, so cyclicity removes all pure powers. Every remaining monomial is supported on exactly two variables. Cyclicity therefore gives $f = \Phi(q), \quad q(X, Y) = f(X, Y, 0)$. The polynomial q is alternating and vanishes on the axes. Restriction of the norm to $z = 0$ gives $N_2 q = 0$.

It remains to prove separate evenness. Put $r(u, v) = q(u, v) - q(-u, v)$. Expanding $N\Phi(q)$ and using alternation and $N_2 q = 0$ gives

$$r(y, z) - r(x, z) + r(x - y, z) = 0. \tag{7.1}$$

Indeed, after removing the three terms of $N_2 q(x, y)$, the six remaining terms are

$$q(y, z) - q(x, z) + q(x - y, z) - q(-y, z) + q(-x, z) - q(y - x, z),$$

which is the left side of (7.1). Thus, we have $r(u + v, z) = r(u, z) + r(v, z)$. Differentiating in v at $v = 0$ shows that $\partial_u r(u, z)$ is independent of u. Since $r(0, z) = 0$, we have $r(u, z) = uh(z)$ for a polynomial h.

Since q is alternating and homogeneous of even total degree, we have $r(v, u) = -r(u, v)$. On the diagonal, this means that $2uh(u) = 0$, so $h = 0$ in characteristic zero. Hence, we have $q(-u, v) = q(u, v)$. Alternation gives evenness in the other variable. Conversely, for a binary polynomial with these properties, the three terms of $N_2 q$ sum to zero and the six terms displayed above cancel. Hence $\Phi(q)$ is a solution. The axis normalization recovers q on $z = 0$. □

Proof of Corollary 1.3. The degree zero spaces are zero. The relation $q(X,Y)+q(-Y,X)=0$, applied twice, gives $q(-X,-Y)=q(X,Y)$. Thus $\mathscr{P}_n^0=0$ when n is odd, as is W_n by Theorem 1.1.

Let $n>0$ be even. The involution $q(X,Y)\mapsto q(-X,Y)$ preserves the binary period relations [17, Section 2, (2.2)–(2.4)]. To match conventions, homogenize by $P(X)\mapsto Y^nP(X/Y)$. The matrix U^2 in that reference gives our substitution $(X,Y)\mapsto(-Y,X-Y)$. Since $U^3=-I$ and $-I$ acts trivially in even degree, the two three-term norms agree. The involution also preserves the axis conditions, so $(q(X,Y)\pm q(-X,Y))/2$ gives the two normalized parity projections.

This is a statement about binary periods, not coordinate sign projections of the ternary norm. The two-term relation makes the odd part symmetric and separately odd, and the even part alternating and separately even. In the notation of Section 2, we have $\mathscr{P}_n^{0,\mathrm{odd}}=\mathcal{B}_n$ and $\mathscr{P}_n^{0,\mathrm{ev}}=\mathcal{B}_n^0$. Theorem 1.2 and Lemma 7.1 identify these spaces with the two sectors exhausting W_n by Lemma 2.1. Both inverse maps are Φ. Since the summands vanish on the axes, its restriction to $z=0$ is exactly q. This proves the rational isomorphism and uniqueness. Compatibility with restriction and Φ in Proposition 6.1 extends the result to F. □

7.2. **Cusp form periods and dimensions.** Let $n>0$ be even, $k=n+2$, and $h\in S_k$. Expanding (1.8) and using the Mellin transform, with the normalization of $\Lambda(h,s)$ in Section 2.3, gives

$$r_h(X,Y)=\sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}i^{j+1}\Lambda(h,j+1)X^{n-j}Y^j. \tag{7.2}$$

Thus the coefficients are explicit multiples of the completed L-values at the critical integers $1,\dots,k-1$ [12, Section 5.10].

Define the even and odd parts in X by

$$r_h^{\mathrm{ev}}(X,Y)=\frac{r_h(X,Y)+r_h(-X,Y)}{2},$$
$$r_h^{\mathrm{odd}}(X,Y)=\frac{r_h(X,Y)-r_h(-X,Y)}{2}.$$

Both satisfy the period relations. The odd part is odd in both variables and vanishes on the axes. The even part is normalized uniquely by

$$r_h^{\mathrm{ev},0}(X,Y)=r_h^{\mathrm{ev}}(X,Y)-r_h^{\mathrm{ev}}(1,0)(X^n-Y^n). \tag{7.3}$$

This subtraction preserves the period relations. Hence r_h^{odd} and $r_h^{\mathrm{ev},0}$ belong to $\mathscr{P}_n^{0,\mathrm{odd}}(\mathbb{C})$ and $\mathscr{P}_n^{0,\mathrm{ev}}(\mathbb{C})$, respectively.

Proof of Corollary 1.4. Eichler–Shimura identifies cusp forms with the two period spaces modulo coboundaries [17, Theorem 2.1]. At level one the coboundary space is spanned by X^n-Y^n. It lies in the even part, and (7.3) selects its unique axis-normalized representative. Thus the two period maps give

$$S_{n+2}\xrightarrow{\sim}\mathscr{P}_n^{0,\mathrm{odd}}\otimes\mathbb{C},\quad h\mapsto r_h^{\mathrm{odd}},$$
$$S_{n+2}\xrightarrow{\sim}\mathscr{P}_n^{0,\mathrm{ev}}\otimes\mathbb{C},\quad h\mapsto r_h^{\mathrm{ev},0}.$$

Composing with Φ proves the claimed identifications. The coefficient-extension proposition transfers the complex dimensions to every characteristic zero field. Odd degrees and degree zero vanish by the classification. Finally, the graded space $S_*:=\bigoplus_{k\geq 0}S_k$ satisfies $S_*=\Delta_{\mathrm{mod}}\mathbb{C}[E_4,E_6]$, where E_4,E_6 are the normalized Eisenstein series of weights four and six and Δ_{mod} is the weight twelve discriminant cusp form. This gives $\sum_k \dim S_k\, t^k=t^{12}/((1-t^4)(1-t^6))$ [19, Chapter VII]. Shift k to $n+2$ and multiply by two to obtain (1.10). □

7.3. **The exact common divisor.**

Proof of Corollary 1.5. By the classification, each nonzero sector is $\Phi(q)$ for a normalized binary period of one of the two parities, and in both cases we have

$$q(u,u)=q(u,-u)=0,\qquad q(-u,v)=-q(v,u).$$

For the symmetric case the first diagonal vanishing follows by evaluating the binary norm at (u, u) and using the axis values. For the alternating case it follows directly from alternation. The other identities follow from the parity conditions. Thus, we have $\Phi(q)(u, -u, v) = 0$. Cyclicity supplies the factors $x + y$, $y + z$, and $z + x$. The construction of $L(q)$ in (2.5) already proves divisibility by $x + y + z$ for either symmetry. The four linear factors are pairwise nonassociate, so their product D divides each sector and every solution.

To prove that D is the entire common divisor, use the degree ten basis $f_{\mathrm{s}} = \Phi(q_{\mathrm{s}})$ and $f_{\mathrm{a}} = \Phi(q_{\mathrm{a}})$ from the introduction. Expansion gives

$$f_{\mathrm{a}} = D(x, y, z)(x - y)(x - z)(y - z) \cdot (x - y - z)(x - y + z)(x + y - z). \tag{7.4}$$

All ten linear factors are distinct. On the other hand, we have

$$f_{\mathrm{s}}(u, u, v) = 2q_{\mathrm{s}}(u, v), \qquad f_{\mathrm{s}}(u + v, u, v) = 2q_{\mathrm{s}}(u, v).$$

Neither is identically zero, and symmetry of f_{s} shows that none of the six additional factors in (7.4) divides it. Therefore, we have $\gcd(f_{\mathrm{s}}, f_{\mathrm{a}}) = D$ up to a nonzero constant. Since both are solutions and D divides every solution, the result follows. □

7.4. **Changing the coefficient ring.** For a commutative $\mathbb{Q}$-algebra R, Proposition 6.1 extends the classification, odd degree vanishing, and divisibility by D to R, including rings with nonzero nilpotents.

More generally, a commutative ring R with torsion free additive group embeds in $R_{\mathbb{Q}} = R \otimes_{\mathbb{Z}} \mathbb{Q}$. A solution's restriction $q(X, Y) = f(X, Y, 0)$ already lies in $R[X, Y]$. The identity $f = \Phi(q)$ and the binary period relations descend coefficientwise from $R_{\mathbb{Q}}$, as do support and odd degree vanishing.

Divisibility by D also descends to R. Its leading coefficient in lexicographic order $x > y > z$ is one, so polynomial division by D can be performed over R without dividing any coefficients. The remainder becomes zero over $R_{\mathbb{Q}}$, and injectivity makes it zero over R as well. The greatest common divisor assertion is made only over fields.

In positive characteristic, the support conclusion fails. Let p be a prime, and let $\overline{f}_{\mathrm{s}}$ be the reduction modulo p of the integral degree ten symmetric solution from the introduction. This reduction is nonzero, since the coefficients of f_{s} include 4 and -25. Define

$$\Pi_p(x, y, z) = \prod_{0 \neq \lambda \in \mathbb{F}_p^3} (\lambda_1 x + \lambda_2 y + \lambda_3 z).$$

Every invertible linear substitution over $\mathbb{F}_p$ permutes the factors, so Π_p is invariant under $\mathrm{GL}_3(\mathbb{F}_p)$. In particular, it is invariant under cyclic permutation and the substitutions in N. Hence

$$N(\Pi_p \overline{f}_{\mathrm{s}}) = \Pi_p N \overline{f}_{\mathrm{s}} = 0,$$

and $\Pi_p \overline{f}_{\mathrm{s}}$ is cyclic. It is a nonzero polynomial of degree $p^3 + 9$. Since the factors of Π_p include x, y, and z, every monomial of this solution involves all three variables. Thus it cannot be a sum of polynomials in two variables. The same example works over every field of characteristic p.

In characteristic two there is a smaller example of degree eleven:

$$g = xyz\,\overline{D}^{\,2} \in \mathbb{F}_2[x, y, z],$$

where $\overline{D}$ denotes the reduction of D. For the integral polynomial $q(X, Y) = X^3Y + XY^3$, direct substitution gives

$$N_2 q(X, Y) = 2(X^2 - XY + Y^2)^2.$$

Modulo two, we have $\overline{D} = \Phi(\overline{q})$ and hence $N\overline{D} = 0$. Moreover, $xyz\,\overline{D}$ is the product of the seven nonzero linear forms in three variables over $\mathbb{F}_2$, so it is invariant under $\mathrm{GL}_3(\mathbb{F}_2)$. Therefore $g = (xyz\,\overline{D})\overline{D}$ is a nonzero solution. It violates both two-variable support and odd degree vanishing.

Appendix A. The restricted apartment presentation

We prove the acyclicity required in Proposition 3.1. Let $X = \mathrm{BA}_3(\mathbb{Z})$ be the partial augmented-frame complex. Its vertices are primitive integral lines $v^{\pm}$, identifying $v, -v$. A frame comes from an integral basis. An augmented frame adds the line of a signed sum of two basis vectors. Their subsets are the simplices of X. Let Y be the full good subcomplex for $\ell = (1, 1, 1)$ modulo three.

Church–Putman [10, Theorem C] prove that X is 2-connected. Deleting vertices can create homology, so we must analyze the deleted faces and fill the resulting obstructions explicitly.

We will use the corresponding rank two fact. The full augmented-frame complex in rank two is the contractible filled Farey complex [10, Remark 1.4]. Two bad vertices cannot be adjacent, since their reductions lie in the same one-dimensional subspace. The link of a bad vertex is therefore a bi-infinite path of good vertices. Its star is a cone on this contractible path, so removing that vertex and the interior of its star preserves the homotopy type. For a map of a sphere and a chosen nullhomotopy, only finitely many simplices are involved. Removing those bad stars successively proves that the good rank two subcomplex is connected and weakly contractible, and hence contractible by Whitehead's theorem [14, Chapter 4].

A.1. **The bad-face filtration.** For an all-bad simplex β, its good link $L_\beta = \mathrm{lk}_X(\beta) \cap Y$ consists of good simplices disjoint from β whose join with β lies in X. Write $\mathrm{or}(\beta)$ for its rational orientation line.

Every simplex is the join of its bad face and its good face. Filter the augmented chain complex of X by the dimension p of the bad face, allowing $p = -1$ for an entirely good simplex. There are no all-bad basis triangles or tetrahedra: their reductions would have to span $\mathbb{F}_3^3$ inside $\ker \ell$. The only all-bad triangles are augmented rank two frames. The filtration therefore has four columns, and gives the spectral sequence

$$E^1_{p,q} = \begin{cases} \widetilde{H}_q(Y; \mathbb{Q}), & p = -1, \\ \bigoplus\limits_{\substack{\beta \text{ all-bad} \\ \dim \beta = p}} \mathrm{or}(\beta) \otimes \widetilde{H}_q(L_\beta; \mathbb{Q}), & 0 \le p \le 2, \end{cases} \quad \Longrightarrow \quad \widetilde{H}_{p+q+1}(X; \mathbb{Q}). \tag{A.1}$$

We use augmented homology, including degree $q = -1$. The differentials have bidegree $d^r : (p, q) \mapsto (p - r, q + r - 1)$.

A chain in the β-summand is a join $\beta * \gamma$, with boundary

$$\partial(\beta * \gamma) = (\partial\beta) * \gamma + (-1)^{p+1} \beta * (\partial\gamma).$$

The second term computes good link homology at fixed p, and the first removes a bad vertex. This gives the E^1-page and its indexing. Boundedness of the filtration and finite support of chains ensure convergence for these infinite complexes [9, Chapter VII, Section 2].

Two types of links are immediate. If a bad augmented triangle spans the saturated plane with basis b, c, and w is an integral complement, its good link is the nonempty discrete set

$$\{(w + rb + sc)^{\pm} : (r, s) \in \mathbb{Z}^2\}.$$

For the bad edge $\{b^{\pm}, c^{\pm}\}$, the same vertices form a square-grid graph, with edges given by differences $\pm b$ and $\pm c$. It is connected, and its first homology is generated by unit squares. We next treat the bad-vertex links.

A.2. **The bad-vertex link.** Fix a bad primitive vector b, and put $L_b = \mathrm{lk}_X(b^{\pm}) \cap Y$. The covector ℓ descends to a nonzero covector on $(\mathbb{Z}^3/\mathbb{Z}b)/3(\mathbb{Z}^3/\mathbb{Z}b)$. Let Z be the good rank two augmented-frame complex in this quotient. It is contractible by the rank two argument above. Projection gives a simplicial map $p : L_b \to Z$.

Over $u \in Z$, the lifts $u_k = u_0 + kb$, $k \in \mathbb{Z}$, of a primitive u_0 form a bi-infinite path P_u with edges $[u_k, u_{k+1}]$. Over $[u, v]$, all cross-edges $[u_k, v_l]$ occur, together with

$$[u_k, u_{k+1}, v_l], \qquad [u_k, v_l, v_{l+1}].$$

Thus this part of the link is the 2-skeleton of the join $P_u * P_v$. Over an augmented base triangle, choose compatible representatives and lifts with $u_0 = u_1 + u_2$. The remaining triangles are

$$\theta_{k,l} = [u_1 + kb, u_2 + lb, u_0 + (k + l)b], \qquad (k, l) \in \mathbb{Z}^2.$$

These exhaust the possibilities: the additive relation of an augmented frame containing b either involves b, giving a path-edge triangle, or relates the other three lines, giving one of the $\theta_{k,l}$.

Lemma A.1. *The link L_b is connected and $H_1(L_b;\mathbb{Q})=0$. Its second homology is generated by the tetrahedral boundaries*

$$\partial[u_k,u_{k+1},v_l,v_{l+1}]$$

over base edges, and by the prism boundaries joining $\theta_{k,l}$ to $\theta_{k+1,l}$ or to $\theta_{k,l+1}$ over augmented base triangles.

Proof. Attach a 3-cell along each displayed sphere, obtaining L_b^+. Filter this complex by the dimension of its image in Z:

$$F_{-1}=\varnothing\subset F_0\subset F_1\subset F_2=L_b^+.$$

We compute the homology of each successive pair.

Over a vertex, the path P_u is contractible. Over an edge $[u,v]$, the tetrahedral cells complete the join P_u*P_v. Relative to the endpoint fibers, the contribution is

$$H_i(P_u*P_v,P_u\sqcup P_v;\mathbb{Q})=\begin{cases}\mathbb{Q}, & i=1,\\ 0, & i\neq 1.\end{cases}$$

This follows from the long exact sequence of the pair: the join is contractible and the two endpoint fibers are contractible components. Any cross-edge from P_u to P_v represents the generator.

Over an augmented base triangle, the relative complex has only the coherent triangles and the prism cells. It is

$$\mathbb{Q}[\mathcal{E}(\mathbb{Z}^2)]\xrightarrow{\delta}\mathbb{Q}[\mathbb{Z}^2],\qquad \delta[\nu,\nu']=\theta_{\nu'}-\theta_{\nu},\tag{A.2}$$

in degrees three and two, where $\mathcal{E}(\mathbb{Z}^2)$ consists of the oriented horizontal and vertical edges of the integer grid. The side faces disappear in the relative quotient. Connectedness of the grid makes $\operatorname{coker}\delta=\mathbb{Q}$, while its kernel contributes only in degree three.

Consequently, through total degree two the spectral sequence $E^1_{p,q}=H_{p+q}(F_p,F_{p-1};\mathbb{Q})$ is concentrated in the row $q=0$. This row is the simplicial chain complex of Z: cross-edges have the expected endpoints, and coherent triangles have the expected three boundary edges. The only additional term, $E^1_{2,1}$, has zero possible targets $E^1_{1,1}$ and $E^2_{0,2}$. Since Z is contractible, L_b^+ is connected and has zero homology in degrees one and two.

Only 3-cells were added to L_b. The long exact sequence of (L_b^+,L_b) therefore preserves connectedness and first homology, and gives a surjection

$$H_3(L_b^+,L_b;\mathbb{Q})\longrightarrow H_2(L_b;\mathbb{Q}).$$

The generators on the left map to the attaching spheres, which are precisely the two families in the statement. □

A.3. **The three possible obstructions.** In the good column $p=-1$ of (A.1), the connected bad-vertex and bad-edge links and nonempty bad-triangle links rule out incoming differentials to $\widetilde{H}_0(Y)$. For $H_1(Y)$, the possible sources are

$$H_1(L_b),\qquad \widetilde{H}_0(L_e),\qquad \widetilde{H}_{-1}(L_\tau),$$

for bad vertices b, edges e, and triangles τ. All vanish by the link calculations.

For $H_2(Y)$, the possible sources are $H_2(L_b)$ on the first page, $H_1(L_e)$ on the second, and $\widetilde{H}_0(L_\tau)$ on the third. On later pages one takes their surviving subquotients. We show that each incoming differential is zero on the generators described above. Brackets below denote oriented simplices of primitive lines.

The first differential. Let (b,c,w) be an integral basis with b,c bad and w good. Put

$$A=w,\quad B=w+b,\quad C=w+c,\quad D=w+b+c,\qquad W=A+D=B+C.$$

All five vectors are good: if $\ell(w)=1$ modulo three, then $\ell(W)=2$. The all-good augmented-frame chain

$$[W,B,C,D]-[W,A,C,D]+[W,A,B,D]-[W,A,B,C]\tag{A.3}$$

has boundary $\partial[A,B,C,D]$. This fills each tetrahedral generator from Lemma A.1. Indeed, normalize the two lifted quotient vectors to have the same nonzero covector value. Their difference is a bad vector c, giving this configuration.

For a prism generator, write

$$Z = X + Y, \qquad X' = X + b, \qquad Z' = Z + b = X' + Y.$$

Its sphere is the boundary of the two all-good augmented tetrahedra

$$[X, Y, Z, X'] + [Y, Z, X', Z']. \tag{A.4}$$

Interchanging X, Y treats the other parameter direction. Hence every generator of $H_2(L_b)$ has zero image in the good column.

The second differential. A generator of $H_1(L_e)$ is a unit square in the good link of a bad edge. In the notation above, its fillings in the two bad-vertex links are

$$[A,B,C] + [B,D,C], \qquad [A,B,D] + [A,D,C].$$

Their difference is $-\partial[A,B,C,D]$, which bounds (A.3) in Y. These fillings therefore give zero incoming second differential.

The third differential. Let $u, v, s = u + v$ be a bad augmented core, let w be a good complement, and put $w' = w + u$ and $q = v - w$. Consider

$$\begin{aligned} Z_0 = \; & -[w,q,v,s] - [w,v,u,s] - [w,u,w',s] + [w,q,v,w'] \\ & + [w,q,w',s] + [w,v,u,w'] + [q,v,w',s] + [v,u,w',s]. \end{aligned} \tag{A.5}$$

Each tetrahedron is an augmented frame, using one of

$$v = w + q, \qquad s = u + v, \qquad w' = w + u, \qquad s = q + w'.$$

Expansion gives $\partial Z_0 = 0$. Its terms with three bad vertices are $-[w,v,u,s]+[v,u,w',s]$, so its highest filtered component represents, up to orientation, the adjacent complement difference $[w']-[w]$. Since this component lifts to the absolute cycle Z_0, all its successive differentials are zero. Translating w and interchanging u, v gives every adjacent difference in the complement lattice. These generate $\widetilde{H}_0(L_\tau)$, so the incoming third differential is zero.

There is no outgoing differential from the good column. All incoming differentials in total degrees at most two have now been excluded. The vanishing of the corresponding homology of X therefore gives

$$\widetilde{H}_0(Y;\mathbb{Q}) = H_1(Y;\mathbb{Q}) = H_2(Y;\mathbb{Q}) = 0.$$

This is the restricted acyclicity needed in Proposition 3.1.

Appendix B. Coefficient comparisons for the symmetric argument

We prove the seed finite-difference identity and identify literal residues with binary periods and their Hecke action. The comparison must preserve individual coefficients: pole removal uses an extreme period.

B.1. **The finite-difference identity.** The left side of (4.9) is $\sum_{i=1}^{12} \epsilon_i (Nf)(T_i)$ with the substitutions in Table 3, read down the left column and then the right. In the thirty-six expanded terms, those from $Af, A^2 f$ cancel in opposite-sign pairs with the same unordered arguments. The twelve terms from f are the three second differences. This proves the identity for every symmetric f with $Nf = 0$, without a homogeneity or degree assumption.

ϵ_i	T_i	ϵ_i	T_i
$+$	$(b, a+c, -c)$	$-$	$(b-c, b, a-b+c)$
$-$	$(b, b+c, a-b-c)$	$+$	$(b, a-c, c)$
$+$	$(c-b, a-b, -c)$	$-$	$(a-b, a-b+c, -a-c)$
$-$	$(a-b-c, a-b, c-a)$	$+$	$(-b-c, a-b, c)$
$+$	$(-a, b-a+c, -c)$	$-$	$(-a-c, -a, b+c)$
$-$	$(-a, -a+c, b-c)$	$+$	$(-a, b-a-c, c)$

TABLE 3. The signed substitutions expressing (4.9) as a sum of three-term norms.

B.2. **The literal residue and its Hecke action.** We compare the literal $1/a$ residue used in Section 4.4 with the binary period coordinate, without mixing coefficients.

Let $H = \{a = 0\}$, let P_H be its stabilizer in $G = \mathrm{GL}_3(\mathbb{Z})$, and put $D_d = \mathbb{Q}[b, c]_d$ with the right substitution action. All rational hyperplanes form one G-orbit, so Shapiro reduces the direct sum residue module to its fiber at H. The top homology of the rank two radical of P_H gives the Hochschild–Serre edge map:

$$\mathrm{Sh} : H_3(G, \mathcal{R}_n \otimes \det) \xrightarrow{\sim} H_3(P_H, \mathcal{R}_{n,H} \otimes \det),$$
$$\kappa : H_3(P_H, \mathcal{R}_{n,H} \otimes \det) \xrightarrow{\sim} H_1(\mathrm{GL}_2(\mathbb{Z}), D_{n+1}).$$

The map res_H reads the literal residue, and edge_2 reads the coefficient of the standard oriented binary well-rounded edge. The next lemma compares them with compatible orientation and averaging conventions.

Lemma B.1. *With compatible orientation and averaging normalizations, the diagram*

$$\begin{array}{ccc} H_3(G, \mathcal{R}_n \otimes \det) & \xrightarrow{\mathrm{Sh}} & H_3(P_H, \mathcal{R}_{n,H} \otimes \det) \\ {\scriptstyle \mathrm{res}_H}\big\downarrow & & \big\downarrow{\scriptstyle \kappa} \\ \mathcal{B}_{n+1} & \xleftarrow{\mathrm{edge}_2} & H_1(\mathrm{GL}_2(\mathbb{Z}), D_{n+1}) \end{array} \tag{B.1}$$

commutes. Changing these normalizations multiplies the comparison by one fixed nonzero scalar, independent of the polynomial. In particular, it does not mix its coefficients. The composite identifies the residue with the usual odd period polynomial in the same variables b, c.

Proof. The stabilizer of $a = 0$ in $\mathrm{GL}_3(\mathbb{Q})$ consists of $g = \left(\begin{smallmatrix} \lambda & 0 \\ u & C \end{smallmatrix}\right)$. On the determinant-twisted residue its action is $q \mapsto \det(C) q(C(b, c))$. Its integral radical $U = \mathbb{Z}^2$ acts trivially on that fiber. Right homology acts on $H_2(U, \mathbb{Q})$ by inverse conjugation, with factor $\lambda^2 / \det C$. The combined factor is therefore λ^2. On the integral Levi this is one. The only total-degree-three Hochschild–Serre term is $(1, 2)$, giving

$$H_3(P_H, \mathcal{R}_{n,H} \otimes \det) \cong H_1(\mathrm{GL}_2(\mathbb{Z}), D_{n+1}).$$

This is κ. Shapiro is Hecke-equivariant for this finite-support system [5, Theorem 3.1]. The following cell calculation specifies the coordinate. The standard binary well-rounded edge has minimum-one Gram matrices $C(t) = \left(\begin{smallmatrix} 1 & t \\ t & 1 \end{smallmatrix}\right)$, $|t| \le 1/2$. A coordinate sign flip reverses the edge, whereas a coordinate swap fixes it. The coefficient is separately odd and symmetric. The three edges meeting at a hexagonal vertex give the binary norm. For a minimum-one binary well-rounded form C and $w \in \mathbb{R}^2$, put

$$\delta_C(w)^2 = \min_{z \in \mathbb{Z}^2} (w - z)^t C (w - z), \qquad G(w, C) = \begin{pmatrix} 1 + w^t C w - \delta_C(w)^2 & w^t C \\ Cw & C \end{pmatrix}.$$

For $C(t)$ the covering radius squared is $1/(2(1 + |t|)) \le 1/2$. For $t \ge 0$ this follows from the Voronoi inequalities $|w_1 + t w_2|, |t w_1 + w_2| \le 1/2$ and $|w_1 - w_2| \le 1$. All vertices have that squared length. A sign flip handles $t \le 0$. Thus the Schur complement is positive.

Translate w into the Voronoi cell of zero. The squared length of an integral vector (k, z) for $G(w, C)$ is $k^2(1 - w^t C w) + (z + kw)^t C (z + kw)$. For $k = 0$ the binary minimum gives the lower bound one. For $k = \pm 1$, the same bound follows from the Voronoi inequality. If $|k| \ge 2$, the first term alone is at least $k^2/2 \ge 2$. The

vector $(1,0)$ and two independent binary shortest vectors have length one, so this is a ternary well-rounded form. For $p = \left(\begin{smallmatrix} \eta & 0 \\ u & D \end{smallmatrix}\right)$, with $\eta = \pm 1$, put $C' = D^t C D$, $w' = D^{-1}(u + \eta w)$. Then a direct computation gives $p^t G(w, C) p = G(w', C')$. The map is injective, and its image is the union of integral parabolic translates of the standard ternary top cell: reduce the binary block and translate w to its Voronoi polygon. Conversely, the inequalities for $(1, z)$ in that cell are exactly the same Voronoi inequalities.

Let W_3 be the ternary retract and W_2 the binary retract. The map $i : (w, C) \mapsto G(w, C)$ identifies this with $\mathbb{R}^2 \times W_2$ and is P_H-equivariant. Both this core and W_3 are contractible. After equivariant subdivision, their augmented cellular chains are rational projective resolutions for P_H: the cell stabilizers are finite and averaging is exact. The cellular inclusion lifts the identity of the trivial module, so the comparison theorem for resolutions identifies its induced map with the canonical group homology map [9, Chapter I, Sections 7–8].

The radical acts trivially on the residue fiber. Quotienting its $\mathbb{R}^2$ coordinate by $\mathbb{Z}^2$ gives the oriented torus that represents $H_2(U, \mathbb{Q})$. Fiber integration on the cellular cross product is $[\mathbb{Z}^2 \backslash \mathbb{R}^2] \otimes (e \otimes q) \longmapsto e \otimes q$, where e is an oriented binary edge and the torus has lattice volume one. Since the Levi has rational cohomological dimension one, bidegree $(1, 2)$ is the only term in total degree three. This fiber map is therefore precisely the Hochschild–Serre map κ.

For the standard binary edge, take w in the Voronoi cell of zero. Then $\delta_C(w)^2 = w^t C w$ and the first diagonal entry of $G(w, C)$ is one. The image is the standard ternary cell, and its residue coefficient at $a = 0$ is the same polynomial $q(b, c)$ throughout the fiber. No derivative or change of the variables b, c occurs in integration. The Jacobian from (w_1, w_2, t) to (Cw, t) is positive. The ternary cell stabilizer has order 48 and the stabilizer also preserving H has order 16. Their three cosets give the three coordinate residues in F_q. Cellular averaging and orientation can consequently change this calculation only by one common nonzero scalar. This proves the commutative diagram after the stated normalization.

Finally, the binary modular symbol from 0 to $i\infty$ crosses the standard edge once. Pairing with an arbitrary dual binary coefficient identifies its edge coefficient with the odd part of $\int_0^{i\infty} h(z)(zc - b)^{n+1}\, dz$, with the same orientation normalization. Thus the comparison is the usual period map in the literal variables [17, Theorem 2.1 and Section 5]. □

The Hecke transfer. For the double coset of $\mathrm{diag}(1, 1, p)$, a complete set of right coset representatives is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ b & c & p \end{pmatrix} \quad (0 \le b, c < p), \qquad \begin{pmatrix} 1 & 0 & 0 \\ a & p & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (0 \le a < p), \qquad \mathrm{diag}(p, 1, 1).$$

These $p^2 + p + 1$ representatives classify index p column lattices by normalizing the last nonzero coordinate of their annihilating covector modulo p.

Every representative preserves the hyperplane $H = \{a = 0\}$, so the Hecke correspondence acts on its residue fiber. After separating the normal coordinate from the two coordinates on H, the first family has, for fixed c, binary block

$$C_c = \begin{pmatrix} 1 & 0 \\ c & p \end{pmatrix}, \qquad 0 \le c < p,$$

while the parameter b occurs only in the unipotent direction. The second family has binary block $C_\infty = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$, with the parameter a again lying in the unipotent direction.

The first two families give p representatives above each binary Hecke coset $C_0, \ldots, C_{p-1}, C_\infty$. The final representative $\mathrm{diag}(p, 1, 1)$ has binary block I and normal scale p.

To compute a fiber contribution, write $t = \mathrm{diag}(\lambda, C)$ and set $U_t = \mathbb{Z}^2 \cap (C/\lambda)\mathbb{Z}^2$ and $e_t = [\mathbb{Z}^2 : U_t]$. Normalize fundamental classes by covolume in the same rational plane. Transfer contributes e_t, inverse conjugation on the radical's top homology contributes $\lambda^2 / \det C$, and the residue coefficient contributes $\det C$. Their product is $e_t \lambda^2$.

For $\lambda = 1$, the radical quotient has p elements, represented by $(0, b)$ for $C = C_c$ and by $(a, 0)$ for $C = C_\infty$. Thus $e_t = p$, and the representatives already constitute the transfer sum. For $\lambda = p$ and $C = I$, one has

$U_t = \mathbb{Z}^2$ and $e_t = 1$, so the contribution is p^2. Therefore, we have

$$T_3(p,1)|_{\mathrm{res}} = p\left(\sum_{c=0}^{p-1}[C_c] + [C_\infty]\right) + p^2[I] = pT_2(p) + p^2 I.$$

Here the brackets denote the usual binary double-coset correspondences. They should not be interpreted as plain substitutions on an arbitrary polynomial. This proves (4.12). Multiplying the literal coordinate by a fixed nonzero scalar does not change this operator relation.

It remains to check that this binary correspondence has the standard analytic Hecke normalization on period polynomials. Put $d = n+1$, and let h be a cusp form of weight $d+2$. For a matrix $g = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$ of positive determinant m, the fractional linear action gives $((gzY - X) \mid g) = \frac{m(zY-X)}{cz+e}$, $\quad d(gz) = \frac{m\,dz}{(cz+e)^2}$. Hence, the differential $h(z)(zY - X)^d\,dz$ transforms with the usual weight $d+2$ Hecke factor. The homological correspondence above is the standard Hecke operator on the period polynomial of h. This is the Hecke-equivariance of the period map in [17, Proposition 5.2, Corollary 5.3, and Remark 5.6].

Appendix C. The analytic boundary comparison

The spectral gap criterion gives the boundary surjection in Proposition 3.4. Convergent sums of cusp form differential forms give the plane–line relation used in Proposition 4.2. These are Eisenstein series constructions. We prove their required constant term identities directly.

Throughout this appendix, calculations with differential forms are made on a neat principal congruence subgroup $\Gamma \subset \mathrm{SL}_3(\mathbb{Z})$, of level divisible by three. Such a subgroup can be chosen normal in both K and $K^\dagger$. Its stabilizer of an integral coordinate flag is a semidirect product of its unipotent lattice and its integral Levi. All rational flags are obtained by integral changes of basis, which preserve the principal congruence subgroup. At the end we take invariants under the finite covering group. The coefficient pairing and all restriction maps commute with this averaging. We use $X_\Gamma = \Gamma\backslash\mathscr{X}$, where $\mathscr{X} = \mathrm{SL}_3(\mathbb{R})/\mathrm{SO}(3)$, and write $\overline{X}_\Gamma$ for its Borel–Serre compactification. The convention is unchanged for a rational conjugate of Γ.

C.1. **A spectral gap and compact support vanishing.** We need compactly supported closed two-forms to be exact in ordinary cohomology. The spectral gap criterion of Bergeron–Venkatesh supplies this vanishing. We explain its application to our noncompact quotient.

Put $G_\mathbb{R} = \mathrm{SL}_3(\mathbb{R})$ and $K_c = \mathrm{SO}(3)$, and let $\mathfrak{p}$ be the space of symmetric trace zero real 3×3 matrices, with complexification $\mathfrak{p}_\mathbb{C}$. Use the invariant metric induced by $\mathrm{tr}(UV)$ on $\mathfrak{p}$. For $E = \mathrm{Sym}^n \mathbb{C}^3$ or its dual, let ρ be its representation and choose the Hermitian metric induced by the standard metric on $\mathbb{C}^3$. The flat-bundle metric at gK_c is $\|v\|_g = \|\rho(g)^{-1}v\|$. Let D be the flat exterior differential, D^* its L^2 adjoint, and $\Delta_j = DD^* + D^*D$ the Laplacian in exterior degree j.

Proposition C.1. *For every $n > 0$ there is a constant $c_n > 0$, independent of Γ, such that every compactly supported smooth E-valued form on X_Γ satisfies*

$$\|D\omega\|^2 + \|D^*\omega\|^2 \geq c_n\|\omega\|^2. \tag{C.1}$$

Consequently the natural map $H^2_c(X_\Gamma, E) \longrightarrow H^2(X_\Gamma, E)$ is zero. The conclusion holds for either determinant twist after descent from a neat normal subgroup.

Proof. The Cartan involution $g \mapsto (g^{-1})^{\mathsf{t}}$ exchanges $\mathrm{Sym}^n \mathbb{C}^3$ with its dual. For $t > 1$, the matrix $\mathrm{diag}(t, t, t^{-2})$ has largest eigenvalues t^n and t^{2n} on these two spaces, respectively. They are therefore nonisomorphic when $n > 0$.

The proof of [7, Lemma 4.1] gives a positive constant ε_E such that

$$\Lambda_{E^*} - \Lambda_\pi \geq \varepsilon_E \quad \text{whenever} \quad \mathrm{Hom}_{K_c}(\textstyle\bigwedge^j \mathfrak{p}_\mathbb{C} \otimes E^*, \pi) \neq 0,$$

for any irreducible unitary representation π of $G_\mathbb{R}$ and any j. Here Λ is the eigenvalue of the Casimir, the invariant second order operator. We use the estimate for unitary representations, not the reference's compact quotient conclusion. Kuga's formula [7, (3.6.1)] identifies the Laplacian on the indicated cochains

with $\Lambda_{E^*} - \Lambda_\pi$, since E and E^* have the same Casimir scalar. The reference uses the Killing metric, which is a positive multiple of ours, so rescaling changes only the positive constant.

Now identify square-integrable forms with

$$L^2\Omega^j(X_\Gamma, E) \cong \mathrm{Hom}_{K_c}(\textstyle\bigwedge^j \mathfrak{p}_\mathbb{C} \otimes E^*, L^2(\Gamma\backslash G_\mathbb{R})).$$

The right regular representation is a direct integral of irreducible unitary representations, the continuous analogue of a direct sum. The uniform bound controls every constituent, including the continuous spectrum, and gives (C.1). Taking the smaller constant for E and its dual gives c_n for both.

On the complete quotient X_Γ, choose compactly supported cutoffs $\chi_R \to 1$ with $|d\chi_R| \le C/R$. The commutators with D, D^* are exterior multiplication and contraction by $d\chi_R$. These bounds and local smoothing show that closing the differential on compactly supported forms gives its distributional domain, and likewise for the adjoint. The estimate extends to the closed quadratic form. Its self-adjoint Laplacian Δ_j therefore has a bounded inverse G_j.

The identity $D^2 = 0$ makes D intertwine the adjacent Laplacians and their resolvents on this closed complex. For compactly supported closed ω of degree two, we have $DG_2\omega = G_3 D\omega = 0$, and hence

$$\omega = (DD^* + D^*D)G_2\omega = D(D^*G_2\omega).$$

Local elliptic regularity makes this primitive smooth. It need not have compact support, which is exactly the claimed vanishing of the map to ordinary cohomology. Determinant twists are trivial on SL_3, and finite group averaging descends a primitive from a neat normal cover. □

Poincaré–Lefschetz duality now gives the boundary surjection in Proposition 3.4, for either coefficient system and its dual.

C.2. **Cuspidal face pairings and Stokes's theorem.** We compare restrictions by pairing global classes with dual coefficients. The face pairings sum to zero only after accounting for corner classes. Let P and Q now denote the upper rational parabolic subgroups of block types $(2,1)$ and $(1,2)$, respectively. Thus P stabilizes the plane $\langle e_1, e_2\rangle$ and Q the line $\mathbb{Q}e_1$. Their intersection is the upper triangular group B. For a face of type $R \in \{P, Q\}$, let N_R be its arithmetic unipotent radical and L_R its arithmetic Levi quotient. The radical is a lattice in $\mathbb{R}^2$, and the Levi has a finite index subgroup acting on a modular curve. We use the same letters for rational conjugates at the other cusps.

For a coefficient system E, define $\mathcal{C}_R(E)$ to be the direct sum over faces of type R of the subspaces

$$H^1(L_R, H^1(N_R, E))$$

whose restrictions at all rational cusps of the binary Levi vanish. These are the mixed cuspidal groups defined in Section 4. They inject into degree two face cohomology by Hochschild–Serre, since L_R has rational cohomological dimension one. Write r_R for restriction when all face components are mixed cuspidal.

The evaluation pairing $E \otimes E^* \to \mathbb{C}$ and the oriented faces give perfect pairings

$$\langle\ ,\ \rangle_R : \mathcal{C}_R(E) \times \mathcal{C}_R(E^*) \longrightarrow \mathbb{C}. \tag{C.2}$$

Here is the coefficient calculation. The two-generator Koszul complex for the radical pairs degree one with degree one into its top orientation line. This is Poincaré duality on its compact torus fiber with dual local systems. On the base, the long exact sequence for a compactified modular curve identifies cuspidal cohomology with the image of compactly supported cohomology. Poincaré–Lefschetz duality identifies the transpose of this compact-support-to-ordinary map with the corresponding map for the dual local system. The induced pairing between the two images is therefore nondegenerate: the annihilator of either image is the kernel of the transpose map. In the classical period description this is a nonzero multiple of the Petersson pairing between holomorphic and antiholomorphic classes [17, Theorem 3.3]. Combining the two pairings proves (C.2). The two internal orientation characters multiply to the face orientation. All determinant and reflection actions are retained before taking finite group invariants, so the descended pairing is still perfect.

Lemma C.2. *Suppose $a \in H^2(X_\Gamma, E)$ and $b \in H^2(X_\Gamma, E^*)$ have only mixed cuspidal restrictions on maximal faces and zero restrictions on all flag strata. Then, we have*

$$\langle r_P a, r_P b\rangle_P + \langle r_Q a, r_Q b\rangle_Q = 0. \tag{C.3}$$

The orientations are included in the two pairings.

Proof. Use smooth representatives on $\overline{X}_\Gamma$, whose inclusion of the interior is a homotopy equivalence. The cup product has degree four, and the integral of its restriction over the oriented boundary is zero by Stokes's theorem.

We identify the integral before discarding any corner coordinate. Let F_P, F_Q be the two unions of maximal faces and let F_B be the union of their flag intersections. Use collar thickenings when forming a cover. A de Rham model for the boundary is the mapping cone

$$\Omega^r(F_P;E)\oplus\Omega^r(F_Q;E)\oplus\Omega^{r-1}(F_B;E),\qquad d(\alpha,\beta,u)=(d\alpha,d\beta,\alpha|_{F_B}-\beta|_{F_B}-du).$$

Choose closed face representatives. Their restrictions to a flag collar are exact, since the collar retracts onto the flag and the restricted cohomology classes are zero. Extend a primitive with a cutoff, and subtract its differential. The new face representatives vanish on a smaller collar. After the same change of representative in the mapping cone, its corner coordinate u is closed.

The corresponding corner class is represented on the boundary by $d\chi\wedge u$, with u pulled back from the flag. Choose the same transverse cutoff χ for the two classes. Face–corner products vanish because the face representatives vanish on its support. Corner–corner products vanish because $d\chi\wedge d\chi=0$. Distinct flag collars can be taken disjoint, since there is no triple intersection of maximal faces in rank three. The remaining integral is the sum of the two compactly supported face pairings. Stokes therefore gives (C.3). This argument changes representatives rather than assuming that the original corner cochain is zero. Finite averaging descends the identity. □

C.3. **Closed cusp form seeds and their sums.** We construct global classes by summing closed plane-face forms. The coefficient weight gives convergence without meromorphic continuation. For the remainder of the appendix let

$$E=E_n\otimes_{\mathbb{Q}}\mathbb{C}=\operatorname{Sym}^n((\mathbb{C}^3)^*)\otimes\overset{-1}{\det},\qquad k=n+3>6,$$

with the left action $\rho(g)f(\mathbf{X})=\det(g)^{-1}f(g^{-1}\mathbf{X})$, where $\mathbf{X}=(X,Y,Z)^{\mathsf{t}}$. Plane horospherical coordinates are specified as follows. Write a point of $\mathscr{X}$ as $p\mathrm{SO}(3)$ with

$$p=\begin{pmatrix}A & tu\\ 0 & t\end{pmatrix},\qquad t>0,\quad \det A=t^{-1},\quad u=(u_1,u_2)^{\mathsf{t}},\quad \tau=Ai\in\mathfrak{H}.$$

Right rotation of A leaves u,t,τ unchanged. Here t is the normal height and (u,τ) are plane-face coordinates. Let $\Gamma_P=\Gamma\cap P(\mathbb{Q})$, and let Λ_P be its projection onto the determinant one binary Levi. If $h\in S_k(\Lambda_P)$, set

$$\mathcal{B}=\tau Y-X+(u_1-\tau u_2)Z,\qquad \omega_h=h(\tau)\mathcal{B}^n(du_1-\tau du_2)\wedge d\tau. \tag{C.4}$$

The symbol $\mathcal{B}$ in this subsection is a linear polynomial, not the binary period space $\mathcal{B}_d$. The form is independent of t. Antiholomorphic forms are obtained by complex conjugation, including the coefficient polynomial and differential.

Lemma C.3. *The form ω_h is closed and is equivariant for Γ_P with coefficients E. Its radical class is the binary degree $n+1$ cusp form class, up to a nonzero scalar.*

Proof. Put $\beta=du_1-\tau du_2$. Then, we have

$$d\mathcal{B}=(Y-u_2Z)d\tau+Z\beta,\qquad d\beta=-d\tau\wedge du_2.$$

Each term in $d(h\mathcal{B}^n\beta\wedge d\tau)$ has a repeated factor β or $d\tau$, and is zero. A radical translation replaces u by $u+v$, and its coefficient action replaces (X,Y) by $(X-v_1Z,Y-v_2Z)$, proving equivariance under the lattice.

For the Levi calculation let $g=\operatorname{diag}(A,t)$, $A=\left(\begin{smallmatrix}a & b\\ c & d\end{smallmatrix}\right)$, and $\delta=\det A>0$. We have $u'=Au/t$ and $\tau'=(a\tau+b)/(c\tau+d)$. Direct substitution gives

$$\mathcal{B}(u',\tau',\mathbf{X})=\frac{\delta}{c\tau+d}\mathcal{B}(u,\tau,g^{-1}\mathbf{X}),\quad \beta'=\frac{\delta}{t(c\tau+d)}\beta,\quad d\tau'=\frac{\delta}{(c\tau+d)^2}d\tau.$$

After including the determinant factor in ρ, this becomes

$$g^*\omega_h = \rho(g)\omega_{\delta^{k/2}(h|_kA)}. \tag{C.5}$$

At the neat integral stabilizer, $t = \delta = 1$, and the modular transformation law proves equivariance. The finite reflection projection supplies the corresponding full-level statement.

At $u = 0$, the radical cochain is $(\tau Y - X)^n(1, -\tau)$. In the potential description of plane-radical cohomology, it is the gradient of $-(\tau Y - X)^{n+1}/(n+1)$. This is precisely the required binary coefficient, with its nonzero scalar specified. Eichler–Shimura theory [17, Theorem 2.1] therefore shows that the holomorphic and antiholomorphic seeds span $\mathcal{C}_P(E)$ after including all plane cusps and finite reflections. □

For a datum at one plane cusp, define

$$\mathcal{E}_P(\omega_h) = \sum_{\gamma \in \Gamma_P \backslash \Gamma} \rho(\gamma)^{-1}\gamma^*\omega_h. \tag{C.6}$$

Equivariance removes the choice of coset representative. Rationally conjugate the finitely many other plane cusps to P and sum the same construction over them.

Proposition C.4. *The series* (C.6) *converges normally with every derivative on compact subsets of $\mathscr{X}$. It defines a closed global two-form.*

Proof. At the base point write $\gamma = pk_0$ with p in plane coordinates and $k_0 \in \mathrm{SO}(3)$. The bottom row of p is $(0, 0, t)$, so $t = \|e_3^*\gamma\|$ and $\det A = t^{-1}$. In the coefficient transported pullback, the factor $\rho(p)$ in (C.5) cancels. The remaining compact action has bounded norm. Furthermore,

$$|(h|_kA)(i)| = \left(\frac{\delta}{|ci+d|^2}\right)^{k/2} |h(Ai)| = (\Im Ai)^{k/2}|h(Ai)|$$

is bounded. Equation (C.5) therefore bounds the norm of a summand by $Ct^{-k/2}$. The cosets inject, with a bounded multiplicity depending only on the congruence subgroup, into primitive integral bottom rows. Consequently, we have

$$\sum_{\gamma} \|\text{summand at } \gamma\| \le C' \sum_{0 \neq v \in \mathbb{Z}^3} \|v\|^{-k/2} < \infty,$$

since $k/2 > 3$. On a compact set, the norms of vg and v are uniformly comparable. Rational conjugation changes the lattice by fixed denominators and does not change convergence.

For derivatives, write $k_0 g = p(k_0, g)k_1(k_0, g)$ for $k_0 \in \mathrm{SO}(3)$ and g in a fixed neighborhood of the identity. The parabolic and compact factors, and each fixed derivative of them, are bounded uniformly in k_0. The parabolic character $\delta^{k/2}$ is unchanged. A right derivative therefore produces finitely many binary invariant derivatives of the cusp vector, compact coefficient matrices, and constants from the character. The binary derivatives are compositions of weight raising, weight lowering, and rotation operators. Each is bounded in its normalized weight: this follows from smoothness on the compact part of a fundamental region and from the exponentially convergent Fourier expansion at every cusp. Thus the same summable row majorant applies to every fixed smooth seminorm. Termwise differentiation is legitimate, and Lemma C.3 proves closedness. □

Choose holomorphic and antiholomorphic seeds whose face classes sum to $\xi \in \mathcal{C}_P(E)$, accounting for the factor $-1/(n+1)$ in Lemma C.3. Denote their sum over plane cusps by $\mathcal{E}_P(\xi)$. Its identity constant term represents exactly ξ.

C.4. **Constant terms as cohomological restrictions.** We show that fiber averaging computes actual restriction classes, then unfold the sums. Neither step takes a limit at the boundary.

Restriction and averaging on the fiber. Write $\mathcal{N}_R$ for the real unipotent radical of a maximal parabolic R, $N_R = \Gamma \cap \mathcal{N}_R$ for its lattice, and $\Gamma_R = \Gamma \cap R$. Its boundary model is $e(R) = \mathcal{N}_R \times \mathfrak{H}$. Fixing the positive normal coordinate defines a Γ_R-equivariant map

$$i_{R,t} : e(R) \longrightarrow \mathscr{X}.$$

Since $e(R)$ is contractible and Γ_R acts freely, the quotient $\Gamma_R\backslash e(R)$ is a classifying space. Thus $i_{R,t}$ realizes group restriction. Different positive heights give equivariantly homotopic maps. At a flag use two normal coordinates.

We describe averaging with its coefficient transport. On a unipotent fiber, a flat section has $s(\gamma u) = \rho(\gamma)s(u)$ for $\gamma \in N_R$. Its periodic gauge is $\rho(u)^{-1}s(u)$, and its flat differential is

$$D_{\mathcal{N}} = d + \rho(\theta)\wedge, \qquad \theta = u^{-1}du.$$

Express a form in the left-invariant coframe and average each coefficient over the compact nilmanifold $N_R\backslash\mathcal{N}_R$, with total Haar volume one. This average is a cochain map: each invariant vector field has integral zero on a derivative, while the matrices $\rho(\theta)$ and the structure constants of the coframe are constant. Its image is the finite-dimensional Lie algebra cochain complex with coefficients E. The coefficient transport is part of this averaging operation.

The nilmanifold theorem with unipotent coefficients [15, Theorem 1.1] states that the canonical inclusion ι of Lie algebra cochains induces an isomorphism on fiber cohomology. Its hypotheses hold here: $\mathcal{N}_R$ is simply connected and nilpotent, N_R is a lattice, and the polynomial coefficient representation is unipotent on $\mathcal{N}_R$. The same holds for the dual, and determinant twists are trivial on the radical. Since the averaging map satisfies $\mathrm{Av}\circ\iota = I$, it induces the inverse cohomology isomorphism. Both maps commute with the lattice-preserving Levi action, which preserves normalized Haar measure and transports the coframe and coefficients together. Filtering by base degree gives the comparison for the whole face bundle.

The same theorem applies at a flag, whose radical is the Heisenberg group. Its Lie algebra differential retains $d\theta_{13} = -\theta_{12}\wedge\theta_{23}$, so this application does not replace the flag fiber by a torus.

The averaged pullback along $i_{R,t}$ therefore represents the restriction class, with coefficient transport included.

Unfolding in the coefficient trivialization. The invariant-frame coefficient of a form ω is

$$\widetilde{\omega}(g)(Y_1,\dots,Y_r) = \rho(g)^{-1}\omega_{g\mathrm{SO}(3)}(dL_gY_1,\dots,dL_gY_r), \qquad Y_i \in \mathfrak{p},$$

where L_g is left translation. If ω is Γ-equivariant, then $\widetilde{\omega}(\gamma g) = \widetilde{\omega}(g)$. Moreover the invariant-frame coefficient of (C.6) is simply $\sum_\gamma \widetilde{\omega}_h(\gamma g)$. Constant term integrals below are taken in this trivialization, which agrees with the periodic fiber gauge above after restriction to a face.

For the unfolding, we record congruence conditions prime by prime. Let $\mathbb{Q}_p$ be the p-adic field, $\mathbb{Z}_p$ its integer ring, and $\mathbb{A} = \mathbb{R}\times\prod_p'\mathbb{Q}_p$ the restricted product with respect to $\mathbb{Z}_p$. Use Lebesgue measure at infinity and $\mathrm{vol}(\mathbb{Z}_p) = 1$ at finite primes, so $\mathbb{Q}\backslash\mathbb{A}$ has volume one. This notation packages the congruence coset calculation: rational row classes give global cosets, and the Chinese remainder theorem records their congruence conditions.

At principal level we retain every determinant component. Restriction and unipotent averaging preserve each component. After averaging to the original level, the finite compact determinant contains all local units, and rational lattice reduction gives one component over $\mathbb{Q}$.

An *induced section* ϕ is the coefficient valued cusp datum transforming under the plane parabolic by (C.5). Its real component is the two-form above, and its finite components record the congruence cosets.

Average (C.6) in the invariant-frame trivialization and unfold. The elementary row reductions use the matrices

$$s_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad w_0 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad u(s,t) = \begin{pmatrix} 1 & s & t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The first matrix interchanges the second and third coordinate lines. Two planes are either equal or distinct. A line is either contained in or complementary to the source plane. A full flag has three relative positions. Gaussian elimination respecting these subspaces gives the following complete list. In a zero row the indicated target root is conjugated into the source binary root E_{12}, possibly with a minus sign, which does not affect

its average.

target	representative	contribution or vanishing root
P	I	ξ
P	s_{23}	$E_{13} \longmapsto -E_{12}$
Q	I	$E_{12} \longmapsto E_{12}$
Q	w_0^{-1}	two-root integral
B	$I,\ s_{23},\ w_0^{-1}$	$E_{12},\ E_{13},\ E_{23} \longmapsto \pm E_{12}$

The binary root average is zero by cuspidality. This statement includes the polynomial coefficient: for a binary period form $h(\tau)(\tau Y - X)^d\, d\tau$, the periodic gauge on a cusp with $\tau = u + iv$ replaces the coefficient by $(ivY - X)^d$. Its remaining u-dependence is $h(u + iv)$, whose average over the cusp width is zero. The gradient and contraction representatives of radical cohomology have the same property. Thus none of the zero entries discards a polynomial factor before averaging.

The identity plane cell returns ξ, since radical volume is one. For the complementary line cell, the target radical intersects the conjugated source parabolic radical trivially, leaving the integral

$$(T\phi)(g) = \int_{\mathbb{A}^2} \phi(w_0^{-1} u(s,t) g)\, ds\, dt. \tag{C.7}$$

Equivalently this is the product of its real integral and its finite congruence integrals when the datum is factorizable. The unfolding is absolutely convergent by Proposition C.4. The integrated radical quotient is compact, so Fubini applies. The preceding zero mode argument gives the cohomological identities

$$r_P \mathcal{E}_P(\xi) = \xi, \qquad r_Q \mathcal{E}_P(\xi) = T\xi, \qquad r_B \mathcal{E}_P(\xi) = 0 \tag{C.8}$$

for all $\xi \in \mathcal{C}_P(E)$. Here r_B denotes restriction at each flag. These are cohomological restriction identities at finite height.

It remains to exclude a nonmixed face component. For a maximal face stabilizer $R = N_R \rtimes L_R$, the only terms in total degree two of Hochschild–Serre are $(1,1)$ and $(0,2)$, because L_R has rational cohomological dimension one. Thus there is an exact sequence

$$0 \longrightarrow H^1(L_R, H^1(N_R, E)) \longrightarrow H^2(R, E) \xrightarrow{\mathrm{res}_{N_R}} H^2(N_R, E)^{L_R} \longrightarrow 0.$$

Choose a flag in this face. Its stabilizer contains N_R, so zero restriction at that flag implies zero restriction to N_R. Exactness puts the face class in the mixed term. The restriction of this term to a flag is its binary cusp restriction: the analogous mixed edge map for the binary cusp stabilizer is injective for the same cohomological dimension reason.

The dual series and the boundary relation. We also need the construction for the dual coefficient system. Let J be the antidiagonal permutation matrix and define

$$\jmath(g) = J(g^{-1})^{\mathsf{t}} J.$$

This exchanges P, Q and pulls E back to its dual, after the fixed linear change of coefficient coordinates by J. Apply the preceding construction to $\jmath(\Gamma)$ and pull back. For each $\eta \in \mathcal{C}_Q(E^*)$, it gives a closed global class $\mathcal{E}_Q^\vee(\eta)$ with

$$r_Q \mathcal{E}_Q^\vee(\eta) = \eta, \qquad r_P \mathcal{E}_Q^\vee(\eta) = T^\vee \eta, \qquad r_B \mathcal{E}_Q^\vee(\eta) = 0. \tag{C.9}$$

The normal height exponent and convergence bound are unchanged. Here $T^\vee$ is the second constant term map. We assume neither invertibility nor an adjoint normalization.

Proposition C.5. *Let $n \geq 4$. If a global degree two class with coefficients E has only mixed cuspidal maximal face restrictions (ξ_P, ξ_Q) and zero flag restrictions, then $\xi_Q = T\xi_P$.*

Proof. Pair the class with $\mathcal{E}_Q^\vee(\eta)$ in Lemma C.2, for arbitrary $\eta \in \mathcal{C}_Q(E^*)$:

$$\langle \xi_P, T^\vee \eta \rangle_P + \langle \xi_Q, \eta \rangle_Q = 0.$$

Pair $\mathcal{E}_P(\xi_P)$ with the same dual class:

$$\langle \xi_P, T^\vee \eta \rangle_P + \langle T\xi_P, \eta \rangle_Q = 0.$$

Subtract and use the perfect line pairing (C.2). This proves graph containment without computing the dimension of the global image. □

C.5. **The real constant term entry.** To compute the required real entry, take $h(\tau) = (\tau+i)^{-k}$. This is a local holomorphic lowest rotation weight vector, not a modular form. In the disk coordinate $w = (\tau-i)/(\tau+i)$, write a vector as $(\tau+i)^{-k}F(w)$. The weight k action differentiates to ∂_w and $w^2\partial_w + kw$, up to fixed nonzero normalizations. They send w^m to mw^{m-1} and $(m+k)w^{m+1}$. Thus the constant vector generates all rotation weight vectors $(\tau-i)^m/(\tau+i)^{k+m}$, with no vanishing raising factor.

The integral is equivariant for the binary Levis identified by the block permutation. Its value on the lowest vector therefore determines its values on these one-dimensional rotation weight spaces. Conjugation gives the antiholomorphic component.

Set

$$L = w_0^{-1}u(s,t) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & s & t \end{pmatrix}, \qquad r^2 = 1 + s^2 + t^2.$$

For the Gram matrix LL^{t}, the plane coordinates and transported coefficient in (C.4) are

$$u = (s,t)/r^2, \qquad \tau = \frac{-st + ir}{1+s^2}, \qquad aX + bY + (i/r)Z,$$

where

$$a = \frac{s - it/r}{1+s^2}, \qquad b = -\frac{1 + ist/r}{1+s^2}.$$

Let $v_1 = E_{12}$, $v_2 = E_{13}$ be line-radical tangent vectors and $x_0 = E_{23}$ the horizontal binary tangent. With $\beta = du_1 - \tau du_2$, differentiation gives

$$(\beta \wedge d\tau)(v_2, x_0) = -i/r^3. \tag{C.10}$$

To verify this differentiation, set $G = LL^{\mathsf{t}}$, $R = G_{33}$, $v = G_{\{1,2\},3}$, and $C = G_{\{1,2\},\{1,2\}} - vv^{\mathsf{t}}/R$. Then the plane coordinates are

$$u = v/R, \qquad \Re\tau = C_{12}/C_{22}, \qquad \Im\tau = \sqrt{\det C}/C_{22}.$$

For a tangent matrix V, substitute $\dot{G} = L(V + V^{\mathsf{t}})L^{\mathsf{t}}$ into these formulas. For $V = v_2, x_0$ the resulting derivatives satisfy $d\tau(x_0) = s\, d\tau(v_2) + 1$ and $\beta(x_0) = s\beta(v_2) + t/r^2$. Substitution gives (C.10).

This entry detects the mixed cohomology class. For a line-radical closed pair (α, β_0), its polynomial coordinate is $Z\alpha - Y\beta_0$. Exact pairs have zero contraction. At $X = Z = 0, Y = 1$, this coordinate reads the negative v_2 component. To exclude a contribution in radical degree two, act by $a_v = \operatorname{diag}(e^{2v}, e^{-v}, e^{-v})$. Under the transported action $\rho(a_v)^{-1}a_v^*$, a monomial of X-degree j in radical cohomological degree one has character $e^{(3-n+3j)v}$. The integral (C.7) has character $e^{(6-k)v}$: the source plane character contributes e^{-kv}, while changing its two radical integration coordinates contributes e^{6v}. Since $6-k = 3-n$, the degree-one component has $j = 0$. A radical degree two term would require $6 - n + 3j = 3 - n$, or $j = -1$, and is absent. The binary rotation weight $k = n + 3$ is the extreme polynomial weight $n + 1$ plus the differential weight two. Hence the holomorphic mixed component is one-dimensional for this local datum, and the coefficient just computed determines its scalar.

Relative to the line seed $h(\tau)(\tau Z - Y)^n(\tau\, dv_1 + dv_2) \wedge d\tau$, where dv_i are the covectors dual to the radical coordinates, the scalar is obtained by dividing

$$I_n = -i \int_{\mathbb{R}^2} r^{-3} h(\tau) b^n \, ds\, dt \tag{C.11}$$

by $(-1)^n(2i)^{-k}$, its corresponding value at $\tau = i$. Use $s = \sqrt{r^2-1}\sin\theta$, $t = \sqrt{r^2-1}\cos\theta$, and write $c = \cos\theta$, $v = \sin\theta$. Then, we have

$$\tau = \frac{irc + v}{c - irv}, \qquad b = -\frac{c - iv/r}{c - irv}, \qquad \tau + i = \frac{i(r+1)e^{-i\theta}}{c - irv}, \qquad r^{-3}ds\, dt = r^{-2}dr\, d\theta.$$

The angular integral is

$$\int_0^{2\pi} e^{ik\theta}(c-irv)^3(c-iv/r)^n\,d\theta = \frac{2\pi(r+1)^k}{2^k r^n}.$$

Indeed $c-iav = ((1-a)e^{i\theta}+(1+a)e^{-i\theta})/2$, and $k=n+3$ means that only the negative frequency from every factor contributes to the constant term. The remaining integral is $\int_1^\infty r^{-n-2}\,dr = 1/(n+1)$. Therefore the real entry is

$$c_\infty = -\frac{2\pi i}{n+1} \neq 0. \tag{C.12}$$

Changing orientation or period normalization rescales this nonzero entry. The conjugate component has the conjugate scalar. Reflection projection uses the full twisted coefficient systems, so equivariance preserves nonvanishing without identifying the two untwisted parities.

C.6. **The finite entries and injectivity.** We prove that bad-plane data map injectively to a good line. The prime 3 determines support, and lattice counts at the other primes give the Euler factors. We then combine these with the real entry and reflection signs.

The level three entry. At the prime 3, let

$$K_1 = \{g \in \mathrm{GL}_3(\mathbb{Z}_3) : e_3^* g = e_3^* \pmod 3\}.$$

There are two K_1-orbits on planes and two on lines, according to whether their reductions are annihilated by e_3^*. A plane section supported on the bad-plane orbit is supported on the double coset PK_1. It is determined by its value v_0 at the identity. This value is fixed by the $\mathrm{GL}_2(\mathbb{Z}_3)$ subgroup of $P\cap K_1$. It is therefore zero unless the binary datum has such a fixed vector. Normalize a nonzero section ϕ by $\phi(1)=v_0$, and put

$$g_0 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

This represents a good-line cusp. We have

$$w_0^{-1}u(s,t)g_0 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ t & s & 1 \end{pmatrix} \in PK_1 \quad\Longleftrightarrow\quad s,t \in 3\mathbb{Z}_3.$$

For necessity, its last row must be a scalar multiple of the last row of a K_1 matrix. The last entry is one, so the scalar is a unit and the other two entries must lie in $3\mathbb{Z}_3$. Conversely the displayed matrix itself lies in K_1 when that condition holds. The integral is therefore supported on $(3\mathbb{Z}_3)^2$, its integrand is v_0, and

$$(T_3\phi)(g_0) = \frac{1}{9}v_0. \tag{C.13}$$

Here T_3 denotes the local factor at the place 3 of T, not the rank three Hecke operator $T_3(p,1)$ used in the residue argument.

For even n, the central $-I_2$ kills the odd degree bad-plane binary cohomology. Assume now that n is odd. The matrix $-I_3$ fixes every rational line and plane and acts on E_n by $(-1)^{n+3}=1$. Thus the restricted functional and its connecting class extend to $\langle K^\dagger, -I_3\rangle$, the stabilizer of the projective level covector. At the bad plane, combining $-I_3$ with the binary sign $\mathrm{diag}(-I_2,1)$ supplies the independent normal sign $\mathrm{diag}(I_2,-1)$.

The normal inducing character is therefore trivial on all units at 3, not only on $1+3\mathbb{Z}_3$. It is unramified elsewhere and trivial at the real place. Applying it to a positive rational prime then shows that its value on that prime's uniformizer is one, since every other local value is one. Thus the character is globally trivial. The binary datum is fixed by the full finite maximal compact subgroup, so it has level one and even weight $k=n+3$.

The bad plane and the good line both have full-level binary Levi. Indeed the stabilizer of the good line $\mathbb{Q}e_3$ in $K^\dagger$ is

$$\left\{\begin{pmatrix} D & 0 \\ 3u & 1 \end{pmatrix} : D \in \mathrm{GL}_2(\mathbb{Z}),\ u \in \mathbb{Z}^{1\times 2}\right\},$$

whereas the bad-plane stabilizer has upper block form with the same binary group. Their coefficient systems have different determinant factors, which we retain below.

The unramified factors as lattice sums. We compute each remaining local factor directly from classical Hecke operators. Fix a normalized eigenform h of full level and weight k. Its finite-spherical datum is the lift fixed by $\mathrm{GL}_2(\widehat{\mathbb{Z}})$, where $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$. On $\mathrm{GL}_2^+(\mathbb{R})$ put

$$F_h(g) = (\det g)^{k/2}(ci+d)^{-k}h(gi), \qquad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

For a finite adelic matrix g_f, choose $\gamma \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\gamma g_f \in \mathrm{GL}_2(\widehat{\mathbb{Z}})$ and set $v_h(g_\infty, g_f) = F_h(\gamma g_\infty)$. Such a γ is a rational basis change for the lattice specified by g_f. Its determinant sign can be changed by an element of $\mathrm{GL}_2(\mathbb{Z})$. Two positive determinant choices differ by $\mathrm{SL}_2(\mathbb{Z})$, so this is well defined. Reflections supply the other real component. Positive scalar matrices act trivially, and the even weight gives the same assertion for the negative scalar. Write $v_h|A$ for right translation by A at one finite place. No irreducible local representation is needed for this notation.

Let $p \neq 3$, and for a lattice $\Lambda \subset \mathbb{Z}_p^2$ of index p^r choose a matrix A_Λ whose columns are a basis of Λ. Right $\mathrm{GL}_2(\mathbb{Z}_p)$-invariance makes the basis choice immaterial. The classical Hecke normalization gives

$$\sum_{[\mathbb{Z}_p^2:\Lambda]=p^r} v_h|A_\Lambda = p^{r(1-k/2)}a_h(p^r)v_h. \tag{C.14}$$

Here is the left/right comparison in this normalization. Column lattices have representatives $A = \left(\begin{smallmatrix} a & 0 \\ b & d \end{smallmatrix}\right)$, with $ad = m = p^r$ and $0 \le b < d$. At finite coordinate one, right translation by A changes the real function to $F_h(A^{-1}g)$. For $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, we have

$$mA^{-1} = SA^{\mathsf{t}}S^{-1}.$$

The scalar m and left multiplication by S do not change F_h. The transposes A^{t} run over $\mathcal{H}_m$, so their sum is $m^{1-k/2}T_m$, evaluated at $S^{-1}g$. The eigenvalue equation and $\mathrm{SL}_2(\mathbb{Z})$-invariance remove that last S^{-1}. This proves (C.14) at finite coordinate one, and hence everywhere: a full-level adelic function is determined there by the same rational lattice reduction used to define v_h.

Call Λ *cyclic* if $\mathbb{Z}_p^2/\Lambda$ is cyclic. A lattice of index p^r is noncyclic exactly when it is contained in $p\mathbb{Z}_p^2$. In that case it is $p\Lambda'$ with $[\mathbb{Z}_p^2 : \Lambda'] = p^{r-2}$. Subtracting these lattices in (C.14) gives

$$\sum_{\substack{[\mathbb{Z}_p^2:\Lambda]=p^r \\ \Lambda \text{ cyclic}}} v_h|A_\Lambda = p^{r(1-k/2)}\{a_h(p^r) - p^{k-2}a_h(p^{r-2})\}v_h, \tag{C.15}$$

where $a_h(p^{-1}) = a_h(p^{-2}) = 0$.

We now relate these lattices to the integral. In $L(s,t) = w_0^{-1}u(s,t)$, integral changes of s,t are right $\mathrm{GL}_3(\mathbb{Z}_p)$-operations. For $\max(|s|_p, |t|_p) = p^r$, $r \ge 1$, the primitive pair $p^r(s,t)$ modulo p^r defines the cyclic lattice

$$\Lambda_{s,t} = \{(u,v) \in \mathbb{Z}_p^2 : su + tv \in \mathbb{Z}_p\}.$$

Each such lattice has exactly $(1-p^{-1})p^r$ primitive defining pairs modulo p^r, differing by units. Each pair corresponds to a coset of $\mathbb{Z}_p^2$ in the integration variables and therefore has measure one.

Choose a basis A_Λ with determinant p^r. Lift its columns (u,v) to $(-su-tv, u, v)$ in $\mathbb{Z}_p^3$, and complete them by a vector on which the row $(1,s,t)$ has value p^{-r}. Such a vector exists by Bézout's identity. The resulting matrix k_0 is integral and has determinant one, and

$$L(s,t)k_0 = \begin{pmatrix} A_\Lambda & * \\ 0 & p^{-r} \end{pmatrix}.$$

The plane transformation law (C.5), with the product formula for rational determinants, gives the finite factor $|\det A_\Lambda|_p^{k/2} = p^{-rk/2}$ times $v_h|A_\Lambda$. The normal character is trivial. Thus (C.15) shows that the shell of valuation r contributes

$$(1-p^{-1})p^{-r(k-2)}\{a_h(p^r) - p^{k-2}a_h(p^{r-2})\}v_h.$$

The integral over $\mathbb{Z}_p^2$ contributes v_h.

Put $t_p = p^{-(k-2)}$ and $H_p(T) = \sum_{r\geq 0} a_h(p^r)T^r = (1 - a_h(p)T + p^{k-1}T^2)^{-1}$. Summing the shells gives the scalar

$$\begin{aligned} c_p &= 1 + (1-p^{-1})\{(1-p^{k-2}t_p^2)H_p(t_p) - 1\} \\ &= \frac{1 - a_h(p)t_p/p + p^{k-1}t_p^2/p^2}{1 - a_h(p)t_p + p^{k-1}t_p^2} = \frac{L_p(h,k-2)}{L_p(h,k-1)}. \end{aligned}$$

Lemma 2.5 gives absolute convergence. The Chinese remainder theorem separates the lattice sums into commuting prime power Hecke sums, acting on h by c_p. We may therefore take their product. Combining it with (C.13) and the real integral gives

$$-\frac{2\pi i}{9(n+1)}\frac{L^{(3)}(h,k-2)}{L^{(3)}(h,k-1)} \neq 0. \tag{C.16}$$

The scalar is expressed in the induced section and differential form normalizations fixed above. Only its nonvanishing is used.

Reflection signs and injectivity. The reflection conventions are important here. In binary degree $d = n+1$, the plane coefficient is $\det^{-1}\mathrm{Sym}^d((\mathbb{C}^2)^*)$, whereas the line coefficient has no binary determinant twist. Hence a reflection acts with opposite signs on the two coefficient systems if that twist is suppressed. All maps above retain these determinant factors and are equivariant on the full coefficient systems. At the real place the holomorphic and antiholomorphic scalars are complex conjugates, $-2\pi i/(n+1)$ and $2\pi i/(n+1)$, as required by the reflection. Consequently equivariance survives the finite reflection projection.

The level one cusp space has a simultaneous Hecke eigenbasis [12, Section 5.8]. The correspondence preserves each eigenform and acts on its holomorphic and antiholomorphic components by the nonzero entries above. Distinct eigenvalue systems determine distinct Fourier expansions, so images of normalized eigenforms cannot cancel. Reflection projection commutes with the correspondence. Injectivity thus descends to the arithmetic component required in Proposition 4.2.

Appendix D. The reduced norm kernel

We prove Lemma 5.3 over $k = \mathbb{F}_2$. In $S = k[a,b,c]$, the generators and invariants of Section 5 have weights $1,2,3$ and $4,6,7$, respectively. The point is to obtain a basis valid in every degree, not only in the finite quotient used below.

D.1. **A projector defined by linear substitutions.** Let $\mathcal{W}$ be the six permutation matrices and let

$$\mathcal{U} = \left\{ \begin{pmatrix} 1 & s & t \\ 0 & 1 & u \\ 0 & 0 & 1 \end{pmatrix} : s,t,u \in k \right\}.$$

For a symmetric polynomial g, define

$$\pi g(\mathbf{x}) = \sum_{P\in\mathcal{W}} \sum_{U\in\mathcal{U}} g(UP\mathbf{x}), \qquad \mathbf{x} = (x,y,z)^{\mathsf{t}}.$$

This is a sum of substitutions in characteristic two, not a normalized average. Its value is symmetric, since permuting x,y,z permutes $\mathcal{W}$.

For the remaining projector identities, let $\mathscr{H}$ be the following set of substitutions, in the notation of (5.6):

$$\begin{gathered}(y,z,x),\quad (y,z,x+y),\quad (z,y,x+z),\\ (x,z,y),\quad (x,z,x+y),\quad (x,z,y+z),\\ (x+y,z,x),\quad (x+y,z,y),\quad (x,y,z),\\ (x,y,x+z),\quad (x,y,y+z),\quad (y,x+z,x),\\ (x+z,y,z),\quad (x,y+z,y),\quad (x,y+z,z).\end{gathered}$$

Set $\Theta h = \sum_{\lambda\in\mathscr{H}} h(\lambda)$. Expansion and cancellation of identical triples of arguments, using symmetry of g, give

$$\overline{N}\pi g = 0, \qquad \pi g = g + \Theta\overline{N} g. \tag{D.1}$$

These operator identities hold in every degree. They give $\pi^2 = \pi$ and $\operatorname{im}\pi = \ker\overline{N}$, with π acting identically on that kernel. The product identity (5.2) makes π linear over $R = k[d_4, d_6, d_7]$.

In the notation of Lemma 5.3, direct substitution gives

$$(\pi(a^2b), \pi(ab^2), \pi(b^3), \pi(b^2c), \pi(ac^2), \pi(bc^2), \pi(c^3), \pi(b^5)) = (K_0, K_1, K_2, K_3, K_4, K_5, K_6, K_7). \tag{D.2}$$

In particular, we have $\pi K_j = K_j$.

D.2. **A basis of the invariant quotient.** Put $I = (d_4, d_6, d_7) \subset S$. We use lexicographic monomial order $a > b > c$. Polynomial division and Buchberger's criterion [11, Chapter 2] give the monic Gröbner basis

$$\begin{gathered}a^4 + ac + b^2,\quad a^3b^2 + ac^2 + b^2c,\quad a^3c + a^2b^2 + abc + c^2,\\ a^2b^3 + ab^2c + b^4 + bc^2,\quad a^2bc + ac^2,\quad a^2c^2 + b^4,\\ ab^4 + c^3,\quad ab^3c,\quad abc^2 + b^3c + c^3,\quad ac^3 + b^5,\\ b^6,\quad b^4c + bc^3,\quad b^2c^2,\quad c^4.\end{gathered}$$

Thus a basis of S/I consists of the residue classes of the 28 standard monomials

$$\begin{gathered}1, b, b^2, b^3, b^4, b^5, c, bc, b^2c, b^3c, c^2, bc^2, c^3, bc^3;\\ a, ab, ab^2, ab^3, ac, abc, ab^2c, ac^2;\\ a^2, a^2b, a^2b^2, a^2c; \qquad a^3, a^3b.\end{gathered}$$

Here a standard monomial is one not divisible by the leading monomial of any element of the displayed Gröbner basis.

We need a basis adapted to the projector. Let

$$\begin{aligned}\mathscr{M} = \{&1, a, b, a^2, c, ab, a^3, b^2, ac, bc, ab^2, a^2c,\\ &c^2, b^3, a^2b^2, ac^2, bc^2, b^4, b^3c, bc^3\}.\end{aligned} \tag{D.3}$$

The residue classes of

$$K_0, \dots, K_7, \qquad (1+\pi)m \quad (m \in \mathscr{M})$$

form another basis of S/I. The following normal forms verify the change of basis over $\mathbb{F}_2$. Here NF_I is the remainder on division by the displayed Gröbner basis:

j	$\mathrm{NF}_I(K_j)$	j	$\mathrm{NF}_I(K_j)$
0	$a^2b + ac$	4	$ab^3 + b^2c$
1	$a^3b + a^2c$	5	ab^2c
2	$abc + b^3 + c^2$	6	c^3
3	b^2c	7	b^5

For $m \in \mathscr{M}$, the normal form of $(1+\pi)m$ is m, except in the five cases

m	$\mathrm{NF}_I((1+\pi)m)$
ab^2	$a^3b + a^2c + ab^2$
a^2c	a^3b
b^3	$abc + c^2$
ac^2	$ab^3 + ac^2 + b^2c$
bc^2	$ab^2c + bc^2$.

Applying the substitutions and reducing by the Gröbner basis gives these formulas. They span each weighted degree of S/I. In degree five, for example, they successively give a^3b, a^2c, ab^2, and bc. Degree six gives c^2, abc, b^3, and a^2b^2, and the other degrees are equally triangular. The 28 spanning classes therefore form a basis.

D.3. **Lifting the basis to the polynomial ring.** To avoid assuming algebraic independence of the invariants, begin with the formal polynomial ring $R' = k[u_4, u_6, u_7]$, acting on S by $u_i \mapsto d_i$. The 28 homogeneous lifts above generate S over R': reduce a homogeneous polynomial modulo I, subtract the corresponding linear combination of the lifts, and write the remainder as $d_4s_4 + d_6s_6 + d_7s_7$. Each s_i has smaller weighted degree, so induction applies.

The degrees of the lifts have generating polynomial

$$\frac{(1-t^4)(1-t^6)(1-t^7)}{(1-t)(1-t^2)(1-t^3)}.$$

Consequently the free graded R'-module on these lifts has the same Hilbert series as S. The surjection just constructed is therefore an isomorphism in each degree. In particular, the map $R' \to S$ is injective, the invariants are algebraically independent, and these lifts form an R-basis of S.

The projector is identity on the K_j and zero on $(1+\pi)m$. Taking its image in this R-basis proves

$$\ker \overline{N} = \bigoplus_{j=0}^{7} RK_j,$$

as asserted in Lemma 5.3.

Appendix E. AxiomProver Lean certificate

This paper was written through human–AI collaboration with AxiomProver, an AI system for mathematical research through formal proof under development at Axiom Math. The collaboration included literature search, numerical experimentation, and the proposal and rejection of candidate strategies before the proofs took shape. The authors set the direction, made the mathematical judgments, and take full responsibility for the correctness of the results.

Theorems 1.1 and 1.2 and the essential argument leading to them are formalized unconditionally in Lean 4.35.0. The repository contains the formal statements, proofs, and instructions for checking the formalization:

`https://github.com/AxiomMath/TernaryPoly`

The formalization was developed collaboratively by humans and AxiomProver and audited by humans and AI. It depends on Lean 4.35.0 and Mathlib, with the dependency revisions pinned in the repository's Lake configuration. The proof development is free of admissions and uses only Lean's standard logical axioms. Mathematical and source attribution is retained in the source headers.

Comparator checks the two main theorem statements against a separate challenge containing independently specified definitions. The solution imports the proved targets. Instructions for running Comparator and its independent kernel checks are included in the repository.

Acknowledgements

The authors thank Henrik Bachmann and Jan-Willem van Ittersum for correspondence related to the conjecture, and A. Raghuram for correspondence related to Eisenstein cohomology.

References

[1] P. Abramenko and K. S. Brown, *Buildings: Theory and Applications*, Graduate Texts in Mathematics, vol. 248, Springer, New York, 2008.
[2] C. Alfes, H. Bachmann, K. Bringmann, and J.-W. van Ittersum, *Multiple Zeta Values and Modular Forms*, Oberwolfach Rep. **23** (2026), no. 1, 433–486.
[3] G. Allison, A. Ash, and E. Conrad, *Galois representations, Hecke operators, and the mod-p cohomology of* $\mathrm{GL}(3,\mathbb{Z})$ *with twisted coefficients*, Exp. Math. **7** (1998), no. 4, 361–390.
[4] A. Ash, *Small-dimensional classifying spaces for arithmetic subgroups of general linear groups*, Duke Math. J. **51** (1984), no. 2, 459–468.
[5] A. Ash and D. Doud, *Reducible Galois representations and the homology of* $\mathrm{GL}(3,\mathbb{Z})$, Int. Math. Res. Not. IMRN **2014**, no. 5, 1379–1408. arXiv:1205.3086.
[6] A. Ash and L. Rudolph, *The modular symbol and continued fractions in higher dimensions*, Invent. Math. **55** (1979), no. 3, 241–250.
[7] N. Bergeron and A. Venkatesh, *The asymptotic growth of torsion homology for arithmetic groups*, J. Inst. Math. Jussieu **12** (2013), no. 2, 391–447.
[8] A. Borel and J.-P. Serre, *Corners and arithmetic groups*, Comment. Math. Helv. **48** (1973), 436–491.
[9] K. S. Brown, *Cohomology of Groups*, Graduate Texts in Mathematics, vol. 87, Springer-Verlag, New York, 1982.
[10] T. Church and A. Putman, *The codimension-one cohomology of* $\mathrm{SL}_n\mathbb{Z}$, Geom. Topol. **21** (2017), no. 2, 999–1032. arXiv:1507.06306.
[11] D. A. Cox, J. Little, and D. O'Shea, *Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra*, fourth ed., Undergraduate Texts in Mathematics, Springer, Cham, 2015.
[12] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Graduate Texts in Mathematics, vol. 228, Springer, New York, 2005.
[13] L. E. Dickson, *A fundamental system of invariants of the general modular linear group with a solution of the form problem*, Trans. Amer. Math. Soc. **12** (1911), no. 1, 75–98.
[14] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002.
[15] H. Kasuya, *de Rham and Dolbeault cohomology of solvmanifolds with local systems*, Math. Res. Lett. **21** (2014), no. 4, 781–805.
[16] W. Kohnen and D. Zagier, *Modular forms with rational periods*, in *Modular Forms* (Durham, 1983), R. A. Rankin (ed.), Ellis Horwood, Chichester, 1984, pp. 197–249.
[17] V. Paşol and A. A. Popa, *Modular forms and period polynomials*, Proc. Lond. Math. Soc. (3) **107** (2013), no. 4, 713–743. arXiv:1202.5802v5.
[18] D. Quillen, *Higher algebraic K-theory. I*, in *Algebraic K-theory, I: Higher K-theories*, Lecture Notes in Mathematics, vol. 341, Springer, Berlin, 1973, pp. 85–147.
[19] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics, vol. 7, Springer-Verlag, New York, 1973.
[20] J.-P. Serre, *Trees*, translated by J. Stillwell, Springer-Verlag, Berlin, 1980.
[21] D. Zagier, *Periods of modular forms and Jacobi theta functions*, Invent. Math. **104** (1991), 449–465.

Axiom Math, 124 University Avenue, Palo Alto, CA 94301
Email address: vasily@axiommath.ai

Axiom Math, 124 University Avenue, Palo Alto, CA 94301
Email address: ken@axiommath.ai